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Paleoecological patterns at the Hadar hominin site, Afar Regional State, Ethiopia

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Abstract

Reconstructing paleoecological patterns associated with hominin taxa, such as *Australopithecus afarensis*, is important for understanding possible evolutionary mechanisms involved in extinction and speciation events. It is critical to identify local, regional, or pan-African causal factors because patterns at these different levels may affect separate populations of the same species of hominin in unique ways. Habitat reconstructions of 12 submembers of the Hadar and Busidima formations (~ 3.8–2.35 Ma) are presented here along with faunal differences in these submembers through time. Habitats with medium density tree and bush cover dominated the landscape through much of the earlier time period in the Hadar Formation. The lowermost Sidi Hakoma Member is the most closed habitat. The Denen Dora Member shows the influence of frequent floodplain edaphic grasslands with high abundances of reducin bovids. There is an influx of ungulates in the Kada Hadar Member (~3.2–~2.96 Ma) that indicates a more arid habitat populated by mammals that were recovered from earlier deposits further south in Ethiopia and Kenya. In the younger deposits from the Busidima Formation at Hadar, the landscape was open wooded grassland with some floodplain environments. The fossil assemblages from the Busidima Formation show a substantial species turnover. Although high numbers of *A. afarensis* specimens are associated with the lower Sidi Hakoma Member, they clearly inhabited a variety of habitats throughout the entire Hadar Formation. *Australopithecus afarensis* from Laetoli through Hadar times appears to have been a eurytopic species.
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Keywords: *Australopithecus afarensis*; Paleoecology; Species turnover

Introduction

Paleoecological patterns include habitat reconstructions for fossil assemblages, identification of both habitat and species changes through time and across space, and examination of differences in mammalian adaptations and species composition that are associated with habitat differences. A change in species composition through time, for example, may be caused by migrations of species into a region from elsewhere, or by possible speciation and extinction events. If a variety of patterns are examined in the fossil record, causal relationships between climate, local and regional geology, and other factors can be identified. These patterns thus provide the contextual information necessary for understanding hominin evolution.

Various hypotheses regarding the importance of climate and, thus, habitat change, for hominin evolution (Vrba, 1995; Potts, 1998) have instigated many pan-African, regional, and site-based reconstructions of habitat and/or mammalian species turnover events (Behrensmeyer et al., 1997; Reed, 1998; Bobe and Eck, 2001; Alemseged, 2003; Andrews, 2006). Vrba (1995) illustrated pan-African turnover in bovid species between 3.0–2.0 Ma, but there is also evidence of regional and individual locality changes in species across the 3.0–2.0 Ma time period that do not match the pan-African results (Behrensmeyer et al., 1997; Bobe and Eck, 2001; Alemseged, 2003). These differences in the timing of species turnovers may be related to climatic and subsequent habitat change or alterations to habitats that result from local sources such as tectonic or other geological events. Consequently, it is important to examine habitat and species changes on local and regional levels to help identify the intricate patterns and

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causes of habitat change and their association with faunal change and turnover events. Identification of changes in habitats and patterns of species turnover at particular sites and through geological formations will provide ecological information about hominin species and different populations of the same hominin species (e.g., *Australopithecus afarensis* at Laetoli and Hadar). These patterns may be useful for understanding differences between climatic and other local abiotic causes for apparent habitat shifts. Additionally, these ecological data can be compared across the landscape among sites of the same age to examine regional changes in landscape ecology. Regional patterns can be combined in studies of pan-African biogeographic and ecological patterns. By separating local from regional and pan-African patterns, it may be possible to distinguish species turnover events from migration events.

In this paper, I reconstruct Hadar Pliocene habitats for 12 stratigraphic collection units (submembers) included within the Hadar Formation (Basal, Sidi Hakoma, Denen Dora, and Kada Hadar members) and Busidima Formation (Makaamitalu and Danauli basins). These results are compared with other paleoecological results based on palynology (Bonnefille et al., 2004), isotope analysis (Hailemichael, 2000; Hailemichael et al., 2002), and other types of faunal analyses (Bobe and Eck, 2001) that have been done for the Hadar site. I also examine species changes within these same Hadar units, and compare possible turnovers with habitat changes. Species are also compared with habitats and distribution patterns in order to identify eurytopic and possible endemic species at Hadar. I discuss the habitats reconstructed for Hadar with other *A. afarensis* sites such as Laetoli, Maka, and Koobi Fora. These data provide contextual information on the sequential populations of *A. afarensis* at Hadar, and expand the East African regional pattern northward.

The Hadar hominin site in the Afar Region of Ethiopia has yielded large numbers of *A. afarensis* fossils from sediments ranging in age from $>\sim$3.42 to $\sim$2.90 Ma (Walter et al., 1984; Walter and Aronson, 1993; Walter, 1994; Campisano and Feibel, in press). Four geologic members have been recognized in the Hadar Formation: Basal ($\sim$3.8–3.42 Ma), Sidi Hakoma ($\sim$3.42–3.26 Ma), Denen Dora ($\sim$3.26–3.2 Ma), and Kada Hadar ($<\sim$3.2 Ma; Taieb et al., 1976; Campisano, 2007), each bounded by a radioisotopically data tephra. Dates used here are based on the publications of the original authors, but they have been adjusted to reflect the revised age of the Fish Canyon sanidine that is used as the monitor mineral in $^{40}Ar/^{39}Ar$ age estimates (Campisano and Feibel, in press). The dated tephras, sand units, and other marker beds that have been traced across the research area allow partitioning of these members into smaller stratigraphic units. *Australopithecus afarensis* has not yet been recovered from sediments younger than 2.96 Ma at Hadar, partly because much of the time period between $\sim$2.90 and $\sim$2.4 Ma appears to be missing due to geological disconformities. *Homo* has been recovered from sediments dated to $\sim$2.35 Ma (Kimbel et al., 1996) and other younger fossiliferous sediments have also been discovered.

Geological studies have shown that the much of the Hadar Formation was deposited through fluvial activity of a large meandering river system with associated crevasse channels/splays, deltas, and distributary channels, as well as periodic transgressions of paleolake Hadar located east of the research area (Aronson and Taieb, 1981; Tiercelin, 1986; Campisano and Feibel, in press) possibly related to geological activity or climatic cycles in at least the Kada Hadar Member (Yemane et al., 1996; Yemane, 1997; Campisano and Feibel, in press). Shallow lacustrine trangressions into the region are evident in several stratigraphic intervals: 1) upper Basal Member and lower Sidi Hakoma Member, 2) in the middle of the Sidi Hakoma Member, 3) at the top of the Sidi Hakoma Member through the base of the Denen Dora Member, the section in which the lake was likely most extensive, 4) just above the Kada Hadar Tuff in the lower Kada Hadar Member, and 5) during both the depositions of the Bouroukie Tuff 1 (BKT-1) and BKT-2 tephras (Campisano and Feibel, in press). Channel and overbank flood deposits are interspersed between these lacustrine phases with sand and silt units that yield many of the fossils. These geological data are summarized in Fig. 1. These are, of course, generalized characterizations of a multifaceted distribution of sediments across the landscape in the Hadar region.

Extant African habitats range from primary rain forests to deserts. The amount of rainfall, temperature, sunlight, evapotranspiration, soil type, landscape, and weather patterns/seasonality are thus indicative of these habitats, and habitats can inform on these climatic conditions (Archibold, 1995; O'Brien, 1998; 2006; Andrews, 2006). In the tropical belt, however, the seasonal pattern and the amount of rainfall are the most influential factors for the vegetation physiognomy (O'Brien, 1998; 2006; Hawkins et al., 2003; Field et al., 2005). Different habitats often occur together in mosaic patterns because of changes in soil types, subterranean water, etc. Rivers and lakes can cause much of the mosaicism as they provide subterranean water that alters the vegetation physiognomy close to the water and along the length of rivers (Puhakka et al., 1992; Dodds, 1997). Thus, it is possible to have riverine forests abutting almost desert-like habitats such as at modern Hadar. Identifying ancient habitats will thus provide limited information on these aspects of the climate. Descriptions of the physiognomy of extant African habitats used in this study can be found in White (1983), Reed (1997, 1998), and Reed and Rector (2006).

When compared with habitat changes, species turnover patterns at individual sites may provide information on the eurytopic and stenotopic nature of lineages, including hominins. It would be expected that eurytopic species would be present through time and across the landscape in a variety of habitats. In contrast, stenotopic species may only be recovered in particular habitats and their lineages may exhibit higher diversification rates (Vrba, 1980a). Comparisons of the distribution of species can help to understand migration patterns and possible speciation and extinction events, as much as it is possible in the fossil record.

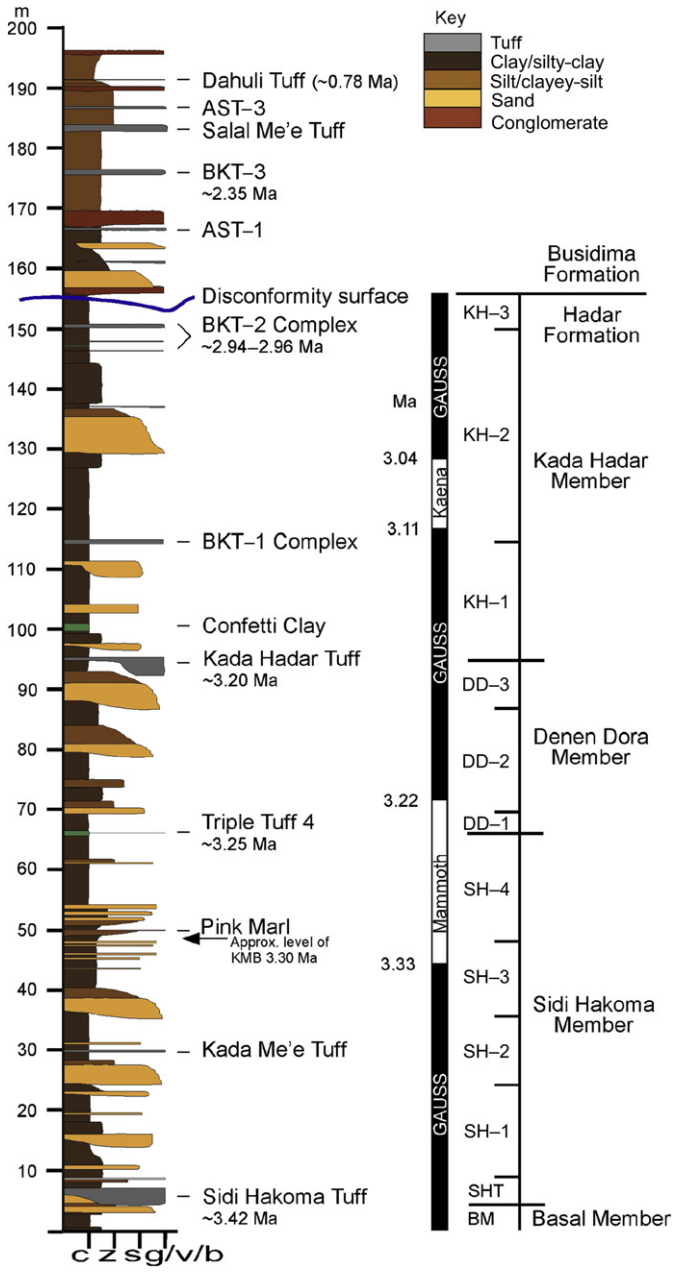


Fig. 1. Composite stratigraphic section of the Hadar and Busidima Formations at Hadar (courtesy of C. Campisano). Tuffs, major marker beds, and submember divisions are labeled alongside the section. $^{40}Ar/^{39}Ar$ dates and paleomagnetic transitions from Schmitt and Nairn, 1984; Renne et al., 1993; Walter and Aronson, 1993; Walter, 1994; Kimbel et al., 1996, 2004; Semaw et al., 1997; and Campisano, 2007. Previously published $^{40}Ar/^{39}Ar$ dates have been recalculated to reflect the updated age of the Fish Canyon sanidine standard (increased by approximately 0.65%; Renne et al., 1998). The left column depicts the changes in depositional environment from clays to gravels. The right hand column equates the submembers discussed in the text with the depositional environments and the dates. Key: c = clays; z = silts; s = sands; g/v/b = gravel, conglomerate, volcanoclastics, bioclastics.

Materials and methods

Comparative sample of extant sites

Forty-two modern African sites consisting of national parks, game reserves, or spatially bounded areas that include habitats from forests through desert were used to establish a baseline of comparative mammalian communities in which habitat structure was known (Swynnerton, 1958; Lamprey, 1962; Child, 1964; Vesey-Fitzgerald, 1964; Sheppe and Osborne, 1971; Smithers, 1971; Rautenbach, 1976, 1978a, 1978b; Ansell, 1978; Behrensmeyer 1991; Delany and Happold, 1979; Bremen and de Wit, 1983; Happold, 1987; Lanjouw, 1987; Emmons et al., 1983; Skinner and Smithers, 1990; Behrensmeyer et al., 1979). Each modern site was assigned a general habitat structure based on observation or written accounts of the area (Table 1). In general, the habitat for each of these sites is mosaic and so the choice of assigning it to one particular habitat or another is somewhat arbitrary. Sites that encompass woodland/bushland/edaphic grassland, for example were assigned to the dominant vegetation type, (e.g., bushland) after the broad categories of White (1983). Parks or reserves that were a combination of disparate vegetation physiognomies were assigned to the ecotone category. For example, the Aberdares National Park includes both forests and mountain heath and is classified here as an ecotone.

The substrate and trophic adaptations for each mammalian species from each extant locality was recorded (Table 2) based on published behavioral observations, examinations of stomach contents, or isotopic studies of enamel, keratin, or hair (Dorst and Dandelot, 1969; Smithers, 1971; Hoffman and Stewart, 1972; Kruuk, 1972; Kingdon, 1974a, 1974b, 1977, 1979, 1982a, 1982b; Happold, 1987; Skinner and Smithers, 1990; Estes, 1991; Kitchener, 1991; Harris and Cerling, 2002; Cerling et al., 2003; Sponheimer et al., 2003). There are 140 large mammal species from all sites and a list of most of these mammals and their trophic and substrate adaptations can be found in Reed (1998).

Ecological diversity analyses have previously used extant mammalian species of all body sizes (e.g., Andrews et al., 1979; Andrews, 1989) and of body mass greater than 500 g (Reed, 1997, 1998). However, the Hadar faunal collection is biased against small to medium-sized mammals. Therefore, a comparative data sample was created from the modern sites by using only mammal species with body masses greater than 4 kg. Soligo and Andrews (2005) have noted that the correct comparative sample should be used for habitat reconstructions otherwise overestimation of fauna with particular adaptations may occur. Of course, the comparative sample of mammalian communities must also be able to inform about habitat type in the modern sites used. Another comparative dataset was created from the modern sites in which only the taxonomic groups most often recovered at Hadar were used. These include: Artiodactyla, Perissodactyla, Proboscidea, and Primates. The primate group included only cercopithecoids as hominins were removed from both modern and fossil sites for that analysis. The selection of adaptations used for this analysis is different from those used with the larger comparative sample (Table 2) and includes a break down of the fruit plus (fruit plus insects or leaves) trophic category into frugivore browsers (fruit and leaf eating; mostly cephalophins and some neotragins), more folivorous primates (colobines), and more omnivorous or frugivorous primates (cercopithecins

Table 1
Extant and fossil localities. Habitats included in the "water" group are indicated by an asterisk (*)

Locality	Code	# of Species > 4 kg	# of species in selected orders	General habitat	Annual rainfall
Aberdares NP, Kenya	Aber	32	21	montane heath/forest (ecotone)	750
Amboseli NP, Kenya	Ambo	40	23	shrubland/bushland	510
Chobe NP, Botswana*	Chob	47	28	scrub woodland/edaphic grassland	650
Congo Basin, DRC	Cong	36	24	edaphic rain forest	1832
East of Cross River, Nigeria	Ecro	24	19	rain forest	1550
East of Niger River, Nigeria	Enig	19	15	rain forest	1596
Gemsbok NP, South Africa	Gems	27	12	shrubland	140
Golden Gate NP, South Africa	Ggat	20	14	grassland/woodland	450
Guinea Woodland, Nigeria	Gwood	43	28	closed woodland	1000
Hadar, Ethiopia	Hada	27	10	shrubland/desert	150
Hluhluwe NP, South Africa	Hluh	43	25	woodland/floodplains	750
Kafue Flats, Zambia*	Kaff	30	22	edaphic grasslands (floodplain)	821
Kafue NP, Zambia*	Kafn	42	24	edaphic grassland (floodplain)/woodland	821
Kapama NP, South Africa	Kapa	40	23	woodland	400
Karoo (Nama), South Africa	Karo	41	25	shrubland	300
Kidepo NP, Uganda	Kide	36	22	shrubland/grassland	625
Kilimanjaro, Tanzania	Kili	16	12	montane forest	1050
Kruger NP, South Africa	Krug	49	30	woodland/bushland	675
Lake Mweru, Zambia*	Lmer	29	24	bushland/woodland	750
Lake Nkuru, Kenya	Lnku	33	18	forest/grassland (ecotone)	1000
Linyanti Swamp, Botswana*	Lswp	29	21	marsh	650
Liwonde NP, Malawi	Liwo	29	19	woodland/edaphic grassland	750
Makokou, Gabon	Mako	28	21	rain forest	1800
Masai Mara, Kenya	Mmar	55	34	forest/grassland (ecotone)	1000
Namib Desert, Namibia	Nami	13	5	desert	<150
Natal Woodland, South Africa	Natw	26	15	closed woodland	875
Nyika NP, Malawi	Nyik	29	17	forest/grassland (ecotone)	1200
Okavango Delta, Botswana*	Okav	44	24	shrubland/woodland/edaphic grassland	600
Rukwa Valley, Tanzania	Rukw	44	28	bushland/woodland	700
Rwenzori NP, Uganda	Rwen	37	21	closed woodland	900
Sahel Savanna, Nigeria	Sahl	23	11	shrubland/edaphic grassland	450
Serengeti Bushland, Tanzania	Serb	52	31	bushland	803
Serengeti NP, Tanzania	Sern	58	36	bushland/woodland/grassland	750
Serengeti Plains, Tanzania	Serp	18	8	grassland	500
SS Grasslands, South Africa	Sgra	26	14	grassland	500
Sudan Woodland, Nigeria	Sudw	37	20	woodland	689
Tai Forest, Ivory Coast	Tai	30	22	rain forest	1900
Tarangire NP, Tanzania	Tara	39	25	scrub woodland	600
Tongwe NP, Tanzania	Tong	29	19	forest/woodland (ecotone)	1012
Tsavo NP, Kenya	Tsav	49	28	bushland/woodland	500
West Lunga NP, Zambia*	Wlun	34	24	bushland/woodland	875
West of Niger River, Nigeria	Wnig	19	14	rain forest	1600
Hadar Formation					
Basal Member and SHT	BS	27	21		
Sidi Hakoma 1	SH-1	27	20		
Sidi Hakoma 2	SH-2	29	21		
Sidi Hakoma 3	SH-3	22	18		
Sidi Hakoma 4	SH-4	21	17		
Denen Dora 1	DD-1	16	14		
Denen Dora 2	DD-2	32	25		
Denen Dora 3	DD-3	28	19		
Kada Hadar 1	KH-1	28	23		
Kada Hadar 2	KH-2	40	31		
Makaamitalu	MAKA	28	23		
Danauli	DANA	28	20		

and papionins) in order to capture differences in faunal representation between each habitat type. This modern collection of data is referred to as the "selected order" database.

These datasets were developed to compare mammalian community adaptations from modern sites in which the general habitat is known with species adaptations of the mammalian fossil assemblages recovered from each of the Hadar fossil submembers. Taphonomic factors that possibly affected the preservation of smaller sized mammals and collection bias that possibly limited the types of species collected at the Hadar

Table 2
Adaptations for extant and fossil mammals

Adaptation	Description	Example
Substrate		
Arboreal		*Colobus guereza* (black and white colobus)
Aquatic		*Hippopotamus amphibious* (hippopotamus)
Fossorial		*Mellivora capensis* (honey badger)
Terrestrial/arboreal		*Panthera pardus* (leopard)
Terrestrial		*Connochaetes taurinus* (wildebeest)
Trophic		
Leaves	Dicot bushes, trees	*Tragelaphus strepsicerus* (kudu)
Meat eater	Flesh	*Panthera leo* (lion)
Meat/bone eater	Flesh, bone	*Crocuta crocuta* (spotted hyena)
Meat/invertebrates	Flesh, insects	*Genetta genetta* (small spotted genet)
Fresh grass	Floodplain/wetland grasses	*Kobus vardoni* (puku)
Fruit plus	Fruit with leaves or insects	*Chlorocebus aethiops* (vervet)
Grass	Grasses	*Alcelaphus bucelaphus* (hartebeest)
Insectivore	Insects	*Orycteropus afer* (aardvark)
Mixed	Mixed grasses and leaves	*Tragelaphus angasi* (nyala)
Omnivore	No preference	*Mellivora capensis* (honey badger)
Roots, tubers	Underground storage organs	*Hystrix africaeaustralis* (porcupine)
Categories for selected orders		
Arboreal		*Colobus guereza* (black and white colobus)
Aquatic		*Hippopotamus amphibious* (hippopotamus)
Terrestrial/arboreal		*Panthera pardus* (leopard)
Terrestrial		*Connochaetes taurinus* (wildebeest)
Leaves	Dicot bushes, trees	*Tragelaphus strepsicerus* (kudu)
Fruit and leaves	Fruits, leaves	*Cephalophus natalensis* (Red duiker)
Grass	Grasses	*Alcelaphus bucelaphus* (hartebeest)
Fresh grass	Floodplain/wetland grasses	*Kobus vardoni* (puku)
Omnivory	No preference	*Potomochoerus laratvus* (bush pig)
Mixed	Mixed grasses and leaves	*Tragelaphus angasi* (nyala)
Colobini	Focus on leaves and seeds	*Colobus guereza* (black and white colobus)
Cercopith	Focus on fruits, leaves, insects	*Papio ursinus* (Chacma baboon)

site are somewhat meliorated by comparing the fossils samples to databases designed to reflect those biases.

Hadar collection units and fossil specimens

The Hadar Formation was divided into ten submembers from the Basal through the upper Kada Hadar members (Gray, 1980; Eck, pers. comm). As mentioned above, these collection units are divided by marker beds, including sand units and tephras, which are not always datable. The Sidi Hakoma Member, for example, ranges in age from ~3.42–~3.26 Ma. It is divided into four submembers, SH-1 (directly above the Sidi Hakoma Tuff dated to ~3.42 Ma), SH-2 (including and above the SH-2 sand), SH-3 (including and above the SH-3 sand), and SH-4 (from extensive gastropod coquinas to Triple Tuff 4 dated to ~3.26 Ma). The Busidima Formation at the Hadar site has not yet been divided into members. Rather, the fossil assemblages analyzed in this study are from drainage basins (Fig. 1).

The Hadar research area has yielded 370 specimens of *A. afarensis,* one specimen of *Homo*, and 7571 additional vertebrate specimens as listed in the Hadar Catalog developed by G. G. Eck. Fossils were collected in the 1970s, 1990s, and 2000s and recorded with stratigraphic position (e.g., DD-3s indicates that the fossil specimen was recovered from the third major sand up from the base of the Denen Dora Member). In some cases, faunal remains were given more than one submember assignment as it was not always possible to tell from which unit a surface-derived fossil eroded. These fossils are not included in the present analysis; only fossil specimens with known provenance (to submember) were used in this study. Therefore, of all of the faunal specimens recovered from the site, 4,158 mammalian specimens were included to derive species lists for each submember (Appendix 1; Table 1). Each submember includes fossils from across the landscape in the Hadar research area. That is, across the approximately 120 km^2 site, all of the fossils from a particular unit are included; this mimics the spatial distribution of animals from West Lunga National Park or the Sahel Savanna, for example. Spatial analysis of the distribution of fossils across the landscape is currently ongoing and is not part of this study (Campisano and Reed, 2007).

For most analyses, the presence of a species rather than its abundance was used. This eliminates some of the taphonomic effects that might be caused by factors affecting species abundances, such as collection practices, carnivore alterations, and other activities. (Behrensmeyer, 1991). Analyses using species abundances in this study, and those discussed in Bobe and Eck (2001), were compared with the analyses utilizing only species presences.

Adaptations of the fossil species were identified using morphological analyses or were based on taxonomic analogy. Although taxonomic analogy is used as a last resort, it has been shown to be effective in identifying trophic or substrate behaviors for ecological diversity analyses (Reed, 1998). Bovid and giraffid mandibular specimens were measured and analyzed morphologically, as in Sponheimer et al. (1999). Bovid dentitions can often be assigned to tribe, but not to species. If a group of mandibles was identified as a particular alcelaphin (e.g., Alcelaphini sp. A) *and* the mandibles were measurable, then the trophic information was included in the analyses (univariate and correspondence analysis) in which trophic and substrate behaviors were required. Carnivore dental specimens were analyzed after Van Valkenburg (1988) and postcranial adaptations after Lewis (1997, pers. comm). Primate information on feeding and locomotor behavior was taken from Jablonski (1993), Elton (2002), Rector (2004), Frost and Delson (2002), and Frost (pers. comm.). However, the postcranial elements for the Hadar primates were not always found in association with taxonomically identifiable cranial elements. Consequently, it was difficult to assign primate postcrania evincing arboreal substrate use to the correct cranially defined taxon (Rector, 2004). Conservatively, only one primate species (*Parapapio* cf. *P. jonesi*) was assigned to this category based on information from Frost and Delson (2002). In fact, more than one of the primate species could have used an arboreal substrate.

Perissodactyl specimens (*Ceratotherium, Diceros, Eurygnathohippus*) were assigned trophic behaviors based on extant species in the case of rhinoceri and based on hypsodonty indices (Janis, 1988) for *Eurygnathohippus*. Artiodactyla, Perissodactyla, and Proboscidea were assigned to the terrestrial substrate use category, except for members of Hippopotamidae, which were classified as aquatic. Harris (1991) suggests that *Hexaprotodon* species were more terrestrial than *Hippopotamus*, and Boisserie (2005) has assigned both Hadar hippotami to ''aff. *Hippopotamus*'' and, therefore, ''aquatic'' seemed the appropriate adaptation to which to assign them.

Suid trophic adaptations are from Harris and Cerling (2002) and Harris et al. (2003). However, while isotope values indicate C_4 plant diets for some fossil suids; this does not necessarily mean that they were grazing. Janis et al. (2002) note that most suid teeth are bunodont and are not indicative of exclusive grazing. Additionally, even extant warthogs, *Phacochoerus aethiopicus* and *P. africanus,* consume many roots and tubers as well as grasses (Kingdon, 1977) and also scavenge carcasses (Bidner, pers. comm). Consequently, assignment of the grazing adaptation to some fossil suids may be somewhat inaccurate as it is based solely on isotopic studies and may not reflect actual diet breadth.

Univariate analysis

The percentages of species exhibiting each adaptation at each modern site (e.g., number of mammal species employing arboreal substrate use divided by the total number of species from that site) were compared by groups of general habitats (Table 1) using a single factor ANOVA. The null hypothesis was that there were no differences in mean proportions of adaptations between habitat categories. These adaptations were further analyzed with the ''Tukey honestly significant difference (HSD) for uneven sample sizes'' post hoc test for differences between pairs of habitats for each adaptation. All habitats with floodplains, swamps and deltas were also tested for significant differences against all other habitats using a one-way ANOVA on the faunal adaptations of aquatic substrate use and fresh grass grazing. Table 1 identifies the habitats included in the ''water'' group with an asterisk. The ANOVA and Tukey HSD analyses were also performed using the selected orders database of modern sites and the mammalian individuals that had the adaptations in the bottom section of Table 2.

Multivariate analyses

Correspondence analysis was used as an exploratory procedure to analyze the relationship among the numbers of species exhibiting particular mammalian adaptations and among the sites. A correspondence analysis was performed on both of the modern site data sets (i.e., data with mammals of body mass larger than 4 kg and data comprising the selected orders of mammals). Each correspondence analysis was repeated incorporating the data from the Hadar submembers. The input matrices for these correspondence analyses thus listed each site (rows) with the numbers of mammal species in each category of adaptation (columns). For example, if there were four species that exhibited arboreal behavior from a particular site, these data were used and not the abundances or percentages of those particular species.

A correspondence analysis transforms the input data to maximize the variance among the rows and columns based on deviations from an expected chi-squared value of the total matrix. This deviation is given in the form of the percentage of inertia on that dimension, the greater the percentage the more meaningful the distribution of points. As much differentiation as possible is placed on the first dimension, and successive dimensions each explain less of the inertia or variation in the data. The sites that plot close together in two-dimensional space are most similar to each other in their patterns of shared numbers of mammalian adaptations. The mammalian adaptations that plot near one another do so because they are associated with the ecological patterns of particular habitats. Although the positions of the sites and positions of the adaptations can be interpreted independently, the distances between the adaptations and the sites cannot be equated. However, general biological statements can be made relating sites and adaptations based on which side of the origins they fall. Using this method, similarities and differences can be identified based on habitat/site and mammalian adaptation groupings. The correspondence analyses were performed using Statistica Version 7.

Species abundances

Minimum numbers of individuals (MNI) were calculated for each bovid tribe and *A. afarensis* based on the taxa recovered at each fossil locality within each submember, again

discounting any recovered specimens without firm provenance. For example, fossils were collected from an Afar Locality (e.g., A.L. 288) and recorded in the catalog with provenance (e.g., DD-1). For each submember, the same species recovered from ten different localities was counted as ten individuals. For the most part, unless the same body part was recovered for the same species at one locality (e.g., two right mandibles), that species was counted as one individual. These data were used in separate analyses to compare to the results obtained through analyses of species presences alone.

A correspondence analysis was used to examine the distribution of the abundances of species recovered in each bovid tribe and of *A. afarensis* for Hadar submembers. In this case, the input matrix was numbers of specimens in each tribe and *A. afarensis* (columns) per submember (rows).

Similarity measures

Species differences between Hadar collection units were also examined. A similarity matrix was created with the Dice/Sorenson Similarity Index (Magurran, 1988), which uses the presence and absence of species in pairs of sites for comparison. This index calculates the variation in species composition between submembers. The equation is

$$Cs = 2j/(a+b)$$

where j is the number of species in common to both submembers, a is the number of species in unit A, and b the number of species in unit B. The shared species are weighted more heavily, thus minimizing the effect of sample size differences. This is especially important in the case of fossil assemblages where the sample sizes may not be indicative of lower diversity but of taphonomic processes. This analysis includes all of the provenanced taxa from the fossil collection units without regard to their use in the other analyses (i.e., if there were mammals smaller than 4 kg in a submember they were included). By the same token, a fossil that could be used in the correspondence analysis (for example, Antilopini species) could not be used in the similarity analysis as it was possibly not the same as an Antilopini species in another level. A cluster analysis was performed on a dissimilarity matrix ($1-C_s$) of the fossil taxa from each stratum to calculate and visually demonstrate the degree of difference among the collection units. The cluster analysis was performed using UPGMA in Statistica Version 7.

Results

Univariate analyses

My previous research using these techniques relied on mammals larger than 500 g, and demonstrated some significant differences in mammalian adaptations among habitats (Reed, 1997, 1998). The dataset of living mammals with body sizes greater than 4 kg used in this study also indicates significant differences in some mammalian adaptations among communities. All adaptations except fresh grass grazing (those that feed in edaphic grasslands) and leaves (browsing) were significantly different between at least one pair of habitats in the ANOVA (Table 3). The Tukey HSD test, however, revealed that habitats with intermediate cover (closed woodlands, bushlands, woodlands, and shrublands) were not significantly different from one another for many of the 16 adaptations. Five adaptations were significantly different among more than six pairs of habitats: arboreality, terrestriality, fruit plus eating, grazing, and mixed feeding (Table 3). Percentages of mammals with the arboreality adaptation differ significantly between forests and all other habitats ($p = 0.0001$) and between closed woodlands and woodlands, shrublands, and grasslands ($p = <0.001$). Percentages of mammals with the arboreal adaptation are also significantly different between bushlands, and shrublands and grasslands ($p = <0.05$) and between ecotones and woodlands, shrublands, and grasslands ($p = 0.002$). The numbers of fruit plus eating mammal species in each habitat type differed significantly between forests and all other habitats ($p = 0.0001$); between closed woodlands and woodlands, shrublands, and grasslands ($p = <0.001$); between bushlands and shrublands and grasslands ($p = <0.02$); and between ecotones and woodlands, shrublands, and grasslands ($p = <0.01$). The arboreal substrate use and fruit plus eating adaptations are best at separating habitats using this database.

Percentages of mammal species with the grazing adaptation are significantly different between mammals with that adaptation in forests and all other habitats except closed woodlands ($p = <0.001$) and between grasslands and closed woodlands, bushlands, shrublands, and ecotones ($p = <0.03$). Adaptations of terrestriality were significantly different between forests and all other habitats and between grasslands and closed woodlands. Finally, the mixed feeding adaptation was significantly different between grasslands and all other habitats. ANOVA results were statistically significant for both aquatic substrate use and fresh grass grazing adaptations between habitats with and without wetland areas (Table 3). Therefore, these seven mammalian adaptations are varyingly successful at separating mammalian communities within certain habitat types. This does not mean that each of these mammalian adaptations successfully separated all the different habitat types. Individual sites may also not have been in the statistical ranges of all mammalian adaptations for their habitat types.

In general, as habitats become more open, arboreal animals decrease and grazing mammals increase (Fernandez-Jalvo et al., 1998). This corresponds directly to abiotic conditions: tree cover is heavier in regions of higher rainfall and better soils (O'Brien, 2006), and where ground water is plentiful (e.g., riverine forests). Arboreal animals are not usually found in shrublands; when they are found, they are found along rivers. Ecotone habitats, such as grassland/forest, have unique signatures, as they have relatively high proportions of species exhibiting arboreal substrate use and fruit plus eating when compared with other adaptations (e.g., Masai Mara and Tongwe National Parks). In forests there are no strictly grazing animals and in open woodlands there are lower proportions of arboreal mammals than are found in ecotones (Table 3). Despite the lack of some significant differences in adaptations among bushlands, woodlands, and shrublands, a general trend from wet and closed

Table 3
ANOVAs and Tukey HSD results from mammalian fauna >4 kg in modern sites. Significant p-values indicated in **bold**; *italics* indicate each adaptation in each habitat type

	SS effect	MS effect	SS error	df error	MS error	F	p
ANOVA of 16 adaptations among modern sites							
Arboreal	0.1653	0.0276	0.0077	34	0.0002	121.4687	**0.00000**
Aquatic	0.0525	0.0088	0.0852	34	0.0025	3.4927	**0.00847**
Fossorial	0.0567	0.0094	0.0502	34	0.0015	6.4018	**0.00014**
Terrestrial	0.3707	0.0618	0.2469	34	0.0073	8.5080	**0.00001**
Terrestrial/arboreal	0.0504	0.0084	0.0687	34	0.0020	4.1521	**0.00311**
Leaves	0.0199	0.0033	0.0547	34	0.0016	2.0635	0.08381
Meat	0.0233	0.0039	0.0344	34	0.0010	3.8389	**0.00498**
Meat/bone	0.0225	0.0037	0.0219	34	0.0006	5.8294	**0.00029**
Fauna/invertebrates	0.0167	0.0028	0.0177	34	0.0005	5.3674	**0.00055**
Fresh grass	0.0217	0.0036	0.0910	34	0.0027	1.3520	0.26190
Fruit and leaves	0.7278	0.1213	0.0507	34	0.0015	81.2886	**0.00000**
Grass	0.1284	0.0214	0.0280	34	0.0008	25.9557	**0.00000**
Insects	0.0285	0.0048	0.0317	34	0.0009	5.1021	**0.00079**
Mixed leaves and grass	0.0635	0.0106	0.0433	34	0.0013	8.3109	**0.00001**
Omnivory	0.0131	0.0022	0.0318	34	0.0009	2.3272	0.05452
Roots and tubers	0.0027	0.0005	0.0049	34	0.0001	3.1575	**0.01433**

	Forests	Closed woodlands	Bushlands	Woodlands	Shrublands	Grasslands	Ecotones
Tukey HSD							
Arboreality means:	*0.1812*	*0.0709*	*0.0338*	*0.0135*	*0.0000*	*0.0000*	*0.0562*
Closed woodlands	**0.0001**						
Bushlands	**0.0001**	0.0656					
Woodlands	**0.0001**	**0.0010**	0.2587				
Shrublands	**0.0001**	**0.0002**	**0.0075**	0.6354			
Grasslands	**0.0001**	**0.0002**	**0.0455**	0.8617	**1.0000**		
Ecotones	**0.0001**	0.8931	0.2487	**0.0015**	**0.0001**	**0.0003**	
Terrestriality means:	*0.4779*	*0.5650*	*0.7147*	*0.7020*	*0.6841*	*0.7955*	*0.6824*
Closed woodlands	0.8680						
Bushlands	**0.0007**	0.3477					
Woodlands	**0.0005**	0.4522	1.0000				
Shrublands	**0.0014**	0.6131	0.9957	0.9997			
Grasslands	**0.0003**	**0.0325**	0.8276	0.7121	0.5265		
Ecotones	**0.0095**	0.6292	0.9964	0.9998	1.0000	0.5080	
Fruit and leaf means:	*0.4533*	*0.2496*	*0.1700*	*0.1044*	*0.0909*	*0.0401*	*0.1992*
Closed woodlands	**0.0001**						
Bushlands	**0.0001**	0.1833					
Woodlands	**0.0001**	**0.0011**	0.0767				
Shrublands	**0.0001**	**0.0004**	**0.0181**	0.9944			
Grasslands	**0.0001**	**0.0001**	**0.0008**	0.2491	0.5183		
Ecotones	**0.0001**	**0.6852**	**0.8912**	**0.0076**	**0.0017**	**0.0002**	
Grazing means:	*0.0000*	*0.0541*	*0.1211*	*0.1339*	*0.1248*	*0.1939*	*0.0884*
Closed woodlands	0.2700						
Bushlands	**0.0001**	0.0928					
Woodlands	**0.0001**	**0.0260**	0.9863				
Shrublands	**0.0001**	0.0652	1.0000	0.9967			
Grasslands	**0.0001**	**0.0001**	**0.0164**	0.0746	**0.0260**		
Ecotones	**0.0006**	0.7622	0.5593	0.1910	0.4321	**0.0003**	
Mixed feeding means:	*0.1283*	*0.0875*	*0.1591*	*0.1370*	*0.1417*	*0.2509*	*0.1132*
Closed woodlands	**0.7979**						
Bushlands	0.7444	0.2064					
Woodlands	0.9993	0.6212	0.9313				
Shrublands	0.9914	0.5179	0.9782	1.0000			
Grasslands	0.0006	**0.0002**	**0.0144**	**0.0014**	**0.0023**		
Ecotones	0.9938	0.9725	0.4145	0.9379	0.8645	0.0002	

	SS effect	MS effect	SS error	df error	MS error	F	p
ANOVA of wetland versus other habitats							
Aquatic substrate use	0.0156	0.0156	0.0233	34.0000	0.0007	22.8059	**0.000034**
Fresh grass grazing	0.0495	0.0495	0.0543	34.0000	0.0016	30.9815	**0.000003**

to dry and open is observed in the five significantly different adaptations (Fig. 2).

Results of the ANOVA and Tukey HSD for the selected orders database are presented in Table 4. All mammalian adaptations except for terrestrial/arboreal substrate use and fresh grass grazing were significantly different between at least one pair of habitats. The Tukey HSD test revealed that most of the percentages of mammalian adaptations within each habitat group were not significantly different among many pairs of habitats, especially among bushlands, woodlands, and shrublands. In fact, most of the differences lie between forests and all other habitats. Thus, despite possibly being a better extant comparison for the mammals recovered from the Hadar submembers, these data are not as good at identifying particular habitat types.

The distribution of the five significantly different adaptations from the Hadar fossil assemblages is detailed in Table 5. The adaptations of the species in each submember at Hadar range in proportions representing closed woodlands to grasslands, but in unique patterns such that a fossil assemblage may exhibit grazing abundances reflecting grassland while the fruit plus eating abundances reflect closed woodland. Hence, a multivariate analysis using these specific adaptations was warranted in order to look at the inter relationship of species with these adaptations and the associations between modern sites and Hadar submembers. The 12 adaptations that were used in the selected orders dataset were all used in a correspondence analysis because none of the adaptations are able to identify distinct intermediate habitats well.

Correspondence analyses and Hadar habitats

The correspondence analysis depicted in Figs. 3 and 4 was derived from the numbers of species exhibiting each of the five adaptations (arboreality, fruit plus eating, grazing, terrestriality, and mixed feeding) that were significantly different between more than one pair of modern habitats plus the aquatic substrate and fresh grass grazing adaptations. The inertia on the first axis is 74.64% and 14.52% on the second axis. Thus, these axes encompass 89% of the variation among the sites and adaptations (Fig. 3). The sites to the left of the x-axis origin (forests, closed woodlands, and ecotones) can be said to be home to more arboreal and fruit plus eating mammals than the sites to the right of the origin (bushlands, woodlands, shrublands, grasslands, and desert), which have more grazers, mixed feeders, and terrestrial mammals. Both the x- and y-axes divide aquatic substrate use and fresh grass grazing (below the y-origin and to the left of the x-origin) from other adaptations which lie closer to the y-origin or above it. The x-axis distribution is essentially a rainfall and seasonal gradient in which the forest sites to the left have high annual rainfall and few seasonal differences in the amount of that rainfall, whereas the sites to the right of the origin have lower mean annual rainfall, which is more seasonal (Foley, 1987). The "water" adaptations separate sites that have abundant water (that is not necessarily dependant on local rainfall, but can be seasonal) for much of the year on the negative side (bottom) of the graph from those sites with less dependable water sources towards the top of the graph (Fig. 4). Thus, sites of overall aridity (from lack of rainfall, annual flooding, or substantial standing water) are located in the top right quadrant. These arid sites include the Namib Desert, Gemsbok, Kidepo, and Tarangire National Parks, and modern Hadar. Sites with seasonal rainfall and significant water for much of the year (wetlands) are in the lower right corner of the graph and include the Linyanti Swamp, Kafue Flats, Chobe, Kafue, and Liwonde National Parks, the Okavango Delta, and Sahel Savanna (Nigeria). West Lunga National Park, Lake

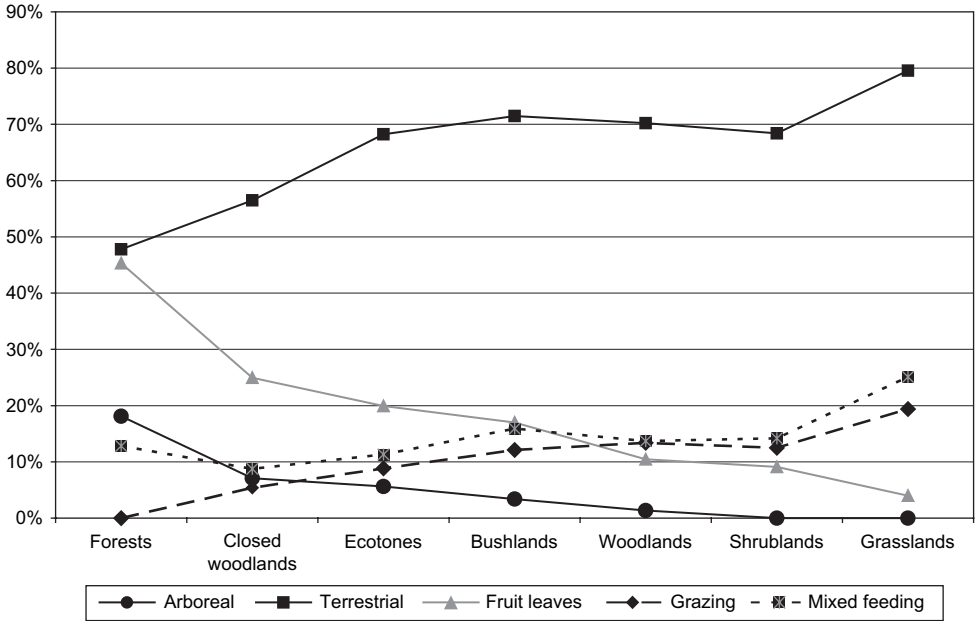


Fig. 2. The five significantly different mammalian adaptations as represented by mean percentages in each type of modern habitat. In general, arboreal substrate use and fruit plus eating decline from forests through grasslands, while grazing and mixed feeding increase. For significant differences between habitat types for each adaptation, see Table 3.

Table 4
ANOVA and Tukey HSD results from mammalian fauna of selected orders in modern sites. Significant p-values indicated in **bold**; *italics* indicate each adaptation in each habitat type

	SS effect	MS effect	SS error	df error	MS error	F	p
ANOVA of 12 adaptations among modern sites							
Arboreal	0.2737	0.0456	0.0245	34.0000	0.0007	63.2311	**0.0000**
Aquatic	0.0663	0.0111	0.1027	34.0000	0.0030	3.6594	**0.0066**
Terrestrial	0.6198	0.1033	0.2071	34.0000	0.0061	16.9602	**0.0000**
Terrestrial/arboreal	0.0128	0.0021	0.0378	34.0000	0.0011	1.9268	**0.1047**
Leaves	0.0776	0.0129	0.1110	34.0000	0.0033	3.9630	**0.0041**
Browsing and fruit	0.2039	0.0340	0.0808	34.0000	0.0024	14.2955	**0.0000**
Grazing	0.4207	0.0701	0.0833	34.0000	0.0024	28.6320	**0.0000**
Mixed feeding	0.1667	0.0278	0.0976	34.0000	0.0029	9.6776	**0.0000**
Fresh grass	0.0304	0.0051	0.1415	34.0000	0.0042	1.2175	0.3215
Omnivore	0.0110	0.0018	0.0128	34.0000	0.0004	4.8741	**0.0011**
Colobines	0.0158	0.0026	0.0204	34.0000	0.0006	4.3887	**0.0022**
Cercopithecines	0.1811	0.0302	0.0983	34.0000	0.0029	10.4424	**0.0000**

	Forests	Closed woodlands	Bushlands	Woodlands	Shrublands	Grasslands	Ecotones
Tukey HSD							
Arboreal means:	*0.2321*	*0.1175*	*0.0458*	*0.0236*	*0.0000*	*0.0000*	*0.0812*
Closed woodland	**0.0003**						
Bushland	**0.0001**	**0.0362**					
Woodland	**0.0001**	**0.0026**	0.7799				
Shrubland	**0.0001**	**0.0002**	0.0751	0.6572			
Grassland	**0.0001**	**0.0002**	0.2249	0.8729	1.0000		
Ecotone	**0.0001**	0.6502	0.3844	**0.0267**	**0.0007**	**0.0026**	
Aquatic means:	*0.1527*	*0.0437*	*0.0543*	*0.0493*	*0.0477*	*0.0406*	*0.0381*
Closed woodland	**0.2177**						
Bushland	**0.0538**	1.0000					
Woodland	**0.0195**	1.0000	1.0000				
Shrubland	**0.0169**	1.0000	1.0000	1.0000			
Grassland	0.0874	1.0000	0.9998	1.0000	1.0000		
Ecotone	**0.0336**	1.0000	0.9991	0.9999	1.0000	1.0000	
Terrestrial means:	*0.5396*	*0.7230*	*0.8296*	*0.8508*	*0.8722*	*0.9237*	*0.7927*
Closed woodland	0.0888						
Bushland	**0.0001**	0.6380					
Woodland	**0.0001**	0.4300	0.9991				
Shrubland	**0.0001**	0.2551	0.9622	0.9985			
Grassland	**0.0001**	**0.0480**	0.6174	0.8375	0.9642		
Ecotone	**0.0003**	0.9260	0.9883	0.8975	0.6766	0.2403	
Browsing means:	*0.0806*	*0.0976*	*0.1283*	*0.1552*	*0.1785*	*0.0406*	*0.1390*
Closed woodland	0.9998						
Bushland	0.7738	0.9941					
Woodland	0.2136	0.8760	0.9819				
Shrubland	**0.0422**	0.5994	0.7306	0.9870			
Grassland	0.9525	0.8805	0.3368	0.0972	0.0254		
Ecotone	0.6748	0.9723	0.9999	0.9993	0.9256	0.2158	
Browsing and fruit means:	*0.2433*	*0.1762*	*0.1276*	*0.0674*	*0.0566*	*0.0292*	*0.1260*
Closed woodland	0.6304						
Bushland	**0.0041**	0.8809					
Woodland	**0.0001**	0.1210	0.3555				
Shrubland	**0.0001**	0.0672	0.1840	0.9996			
Grassland	**0.0001**	0.0125	0.0937	0.9211	0.9840		
Ecotone	**0.0094**	**0.8649**	1.0000	0.4940	0.2967	0.1032	
Grazing means:	*0.0000*	*0.0897*	*0.1921*	*0.2203*	*0.2290*	*0.3604*	*0.1444*
Closed woodland	0.3121						
Bushland	**0.0001**	0.1794					
Woodland	**0.0001**	**0.0394**	0.9533				
Shrubland	**0.0001**	**0.0233**	0.8509	0.9999			
Grassland	**0.0001**	**0.0001**	**0.0007**	**0.0055**	**0.0106**		
Ecotone	**0.0011**	0.8209	0.7302	0.2203	0.1290	**0.0001**	
Mixed feeding means:	*0.1776*	*0.1460*	*0.2387*	*0.2240*	*0.2571*	*0.4010*	*0.2053*
Closed woodland	0.9903						

Table 4 (*continued*)

	Forests	Closed woodlands	Bushlands	Woodlands	Shrublands	Grasslands	Ecotones
Bushland	0.4486	0.3655					
Woodland	0.6717	0.5688	0.9990				
Shrubland	0.1109	0.1780	0.9966	0.9057			
Grassland	**0.0001**	**0.0002**	**0.0026**	**0.0009**	**0.0094**		
Ecotone	0.9814	0.8206	0.9541	0.9978	0.7276	**0.0003**	
Colobine means:	*0.0436*	*0.0556*	*0.0229*	*0.0000*	*0.0000*	*0.0000*	*0.0265*
Closed woodland	0.9965						
Bushland	0.7638	0.6632					
Woodland	**0.0312**	0.1101	0.6714				
Shrubland	**0.0312**	0.1101	0.6714	1.0000			
Grassland	0.1854	0.1101	0.8369	1.0000	1.0000		
Ecotone	0.9231	0.7702	1.0000	0.6135	0.6135	0.7250	
Cercopithecine means:	*0.2651*	*0.1738*	*0.1105*	*0.1061*	*0.1115*	*0.0357*	*0.1687*
Closed woodland	0.3876						
Bushland	**0.0005**	0.7750					
Woodland	**0.0002**	0.7176	1.0000				
Shrubland	**0.0002**	0.7875	1.0000	1.0000			
Grassland	**0.0001**	**0.0485**	0.4537	0.5253	0.4377		
Ecotone	**0.0978**	1.0000	0.6124	0.5303	0.6312	**0.0205**	

Mweru, Rwenzori National Park and the Guinea Woodland (Nigeria) are also below the y-axis origin reflecting the wetter habitats within their mosaics. Woodland and bushland habitats tend to cluster around the origin, with the more xeric habitats (due to lack of rainfall) to the right.

To detect if forests were the cause of the distinct gradation on the placement of the habitats (i.e., were so different from other habitat types that they swamped the data), a second correspondence analysis was performed without forests. All other habitats were in the same relative positions on the graph and, in fact, the axes scales were identical; therefore, this analysis is not figured here.

The Hadar submembers were placed into the correspondence analysis with the modern sites (Fig. 5). Although the forests are removed from the figure in order to see the intermediate habitats better, they were used in the calculations of the correspondence analysis. Hadar habitats were not identical through time as the 12 submembers are distributed across both the x and y origins. There is no directional trend through time in those differences (i.e., from closed and wet to more open and arid). Only SH-1 groups with habitats that indicate less seasonality and higher rainfall than the other submembers. The SH-2, Basal, KH-2, and the DD-1 submembers are in the drier quadrant, but do not group with the driest modern shrubland and desert sites. The other Hadar members are also located in the seasonal rainfall region to the left of the origin, but are in the wetter quadrant. The modern sites have not shifted position due to the introduction of the fossil assemblages indicating that the Hadar submembers are not much different in these seven adaptations compared with these modern habitats.

The results of the correspondence analysis of modern sites using data from the selected orders dataset are shown in Fig. 6. The inertia on both axes accounts for only 72% of the total variation among habitats and although there is a closed to open gradient along the x-axis and a xeric to wet gradient on the y-axis, the results are not as habitat specific as the correspondence analysis using the larger dataset and significantly different adaptations. Some sites group by habitat, but others cluster by the numbers of shared adaptations among the taxa within the communities. For example, Tsavo, Tarangire, and Kidepo National Parks all have six browsing species and are in a quadrant together. In general, all of the sites above the origin have more browsers, more frugivore browsers (cephalophins and neotragins), and/or more colobine monkeys than the sites below. These adaptations were basically combined in the fruit plus eating category in the previous analysis.

Figure 7 shows the Hadar submembers with the modern sites using the selected order sample. Many of the Hadar submembers have an abundance of browsing mammals as well as more colobines than extant habitats. However, the SH-1 collection unit still falls near the ecotones and more closed habitats. Not much can be said about the placement of any of the more intermediate habitats as the modern habitats are also jumbled. It is interesting that a preponderance of browsers appears to be associated with drier habitats in the modern database, although further analyses are required to determine if this is significant for the Hadar reconstructions.

Species abundances

Differences in the numbers of particular bovid tribes reflect the changes observed in habitats from the previous analyses, as well as changes in species dominance over time. Table 6 and Fig. 8 show the distributions of the minimum numbers of individuals for each bovid tribe across the Hadar collection units. Figure 8a shows that there are very few Antilopini bovids until the KH-1 submember, when these bovids increase to 15% and then increase to a maximum of about 27% in the KH-2 submember. Even the deposits above the disconformities (Makaamitalu and Danauli sites) have relatively few antilopins. Although alcelaphins reach their peak in the DD-3 submember, the percentages of both alcelaphin and antilopin

Table 5
Hadar submembers and their habitat classifications for each of the five significantly different adaptations. The actual percentage of mammals exhibiting each behavior in each submember is given in **bold**

Submember	Arboreality	Terrestriality	Fruit and leaves	Grazing	Mixed feeding
Basal	Bushland **4%**	Grassland **78%**	Woodland **11%**	Bushland, woodland, shrubland **11%**	Bushland **18.5%**
SH-1	Bushland **4%**	Bushland, woodland, shrubland ecotone **74%**	Closed woodland **26%**	Bushland, ecotone **4%**	Bushland **18.5%**
SH-2	Bushland, woodland **3%**	Grassland **80%**	Woodland **13%**	Bushland, woodland, shrubland **13%**	Bushland, woodland, ecotone, shrubland **13%**
SH-3	Shrubland, grassland **0%**	Grassland **82%**	Shrubland, grassland **9%**	Bushland, ecotone **9%**	Bushland **18%**
SH-4	Shrubland, grassland **0%**	Bushland, grassland **76%**	Bushland, woodland, ecotone **14%**	Bushland, ecotone **5%**	Bushland **19%**
DD-1	Shrubland, grassland **0%**	Grassland **81%**	Bushland, woodland, ecotone **13%**	Bushland, woodland, shrubland **13%**	Bushland, woodland, ecotone, shrubland **13%**
DD-2	Bushland, woodland **3%**	Bushland, grassland, ecotone **75%**	Shrubland, grassland **9%**	Bushland, woodland, shrubland **13%**	Bushland, woodland, ecotone, shrubland **16%**
DD-3	Shrubland, grassland **0%**	Grassland **82%**	Shrubland, grassland **7%**	Bushland, woodland, shrubland **14%**	Grassland **21%**
KH-1	Shrubland, grassland **0%**	Grassland **83%**	Woodland, shrubland **10%**	Woodland, shrubland **17%**	Grassland **21%**
KH-2	Bushland, woodland **2.5%**	Grassland **83%**	Woodland, shrubland **10%**	Woodland, shrubland **15%**	Bushland **16%**
MAKA	Shrubland, grassland **0%**	Grassland **86%**	Shrubland, grassland **7%**	> Grassland **25%**	Grassland **21%**
DANA	Shrubland, grassland **0%**	Grassland **86%**	Shrubland, grassland **7%**	> Grassland **32%**	Bushland, woodland, ecotone, shrubland **14%**

bovids are high only from the KH-1 submember through the younger Danauli. High proportions of Reduncini species (Fig. 8a) peak within the DD-2 submember. Aepycerotini numbers peak in the DD-1 submember, but otherwise are fairly constant considering taphonomic biases, but they are quite reduced in numbers by ~2.35 Ma (Fig. 8b). Bovini bovids have relatively high percentages in the Basal Member and SH-2 submember, low proportions in the remainder of the Sidi Hakoma, Denen Dora, and Kada Hadar Members, and then increase in the Busidima Formation (Makaamitalu and Danauli). Tragelaphini abundances are similar to the Aepycerotini numbers until the Busidima Formation when they increase, and Aepycerotini decrease. Finally, *Australopithecus* abundance (relative to the bovid tribes) is greatest in the early Sidi Hakoma Member and then drops below 10% in the SH-4 submember and maintains roughly the same proportions throughout the rest of the Hadar Formation deposits.

The correspondence analysis of the numbers of individuals from bovid tribes shows a similar pattern (Fig. 9). The DD-2 submember shares a quadrant with the Reduncini (floodplain and wetlands bovids), the SH-1 submember is in a quadrant with Cephalophini (closed habitat bovids), and the KH-2 submember is most similar to the later sites of Makaamitalu and Danauli in the quadrant with Antilopini and Hippotragini. The other submembers cluster around the origin, as do the bovid tribes of Aepycerotini, Tragelaphini, Bovini, and Alcelaphini. *Australopithecus afarensis* is near this central group as well. This placement in the center of the graph indicates that these species and submembers are relatively similar (i.e., there are no great abundance differences among them).

Faunal similarities in the Hadar submembers

Table 7 shows the numbers of species shared between pairs of collection units. The dendrogram displays the results of the

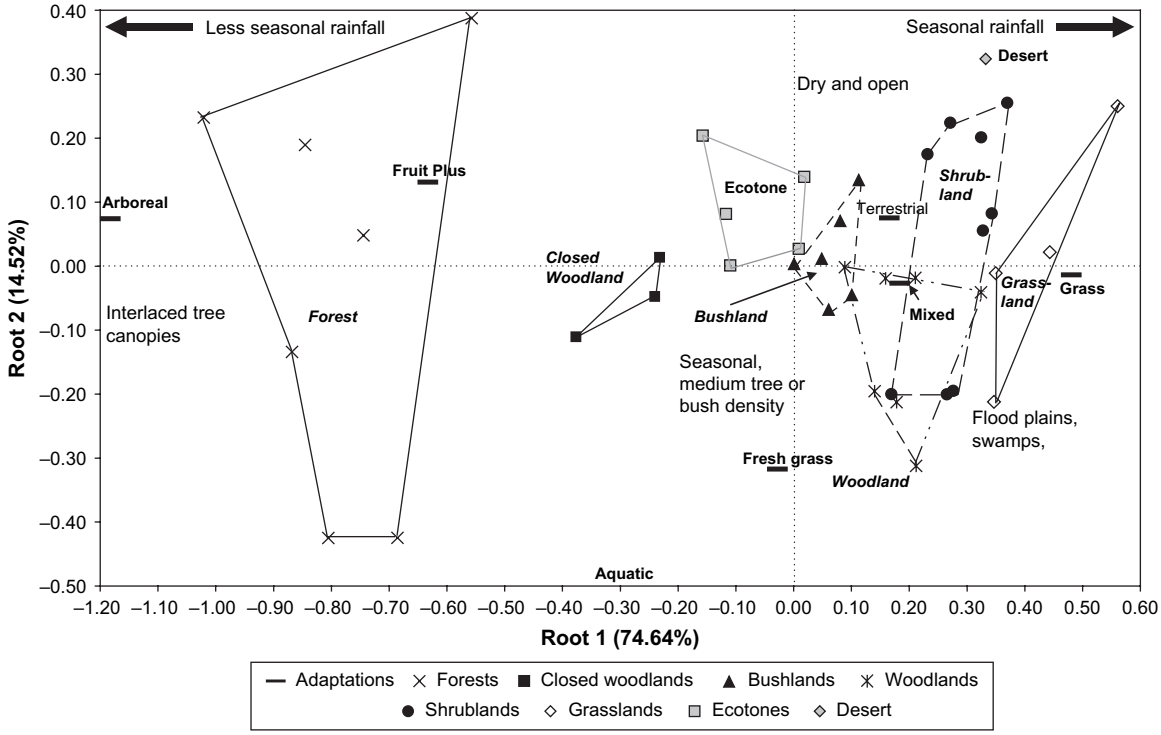


Fig. 3. Correspondence analysis of seven significantly different adaptations and modern habitat types: arboreal, terrestrial, and aquatic substrate use, and fruit plus, fresh grass, mixed, and grass trophic adaptations. The adaptations are underlined with the complete description written above and are fully described in Table 2 with mammalian examples of each. Each modern site within each habitat group is connected with lines to illustrate the area encompassed. Adaptations are positioned from those that are prominent in forests and also closed habitats (arboreal and fruit plus), to adaptations that exist in open habitats (grass) along the x-axis. The only significant adaptations that separate sites on the y-axis are aquatic substrate use and fresh grass grazing. These water-based adaptations are utilized by many species in habitats that encompass edaphic grasslands, swamps, and vleis. The numbers of species that exhibit these seven adaptations in each of the sites also separate the types of habitats on a wet (from rainfall), closed to dry, more open gradient from left to right, and a wet (based on wetlands) to dry gradient from bottom to top on the y-axis. It can clearly be seen that the woodland habitats overlap both bushland and shrubland habitats, although the bushland and shrubland habitats do not overlap with one another. These intermediate density habitats are more difficult to predict precisely using a faunal subset of mammals greater than 4 kg.

dissimilarity matrix created with the Dice-Sorenson Index (Fig. 10). The fauna from *A. afarensis* bearing submembers from the Hadar Formation (Basal through KH-2) group separately from the two areas from the Busidima Formation. Within the *A. afarensis* cluster, the Basal and Sidi Hakoma fossil assemblages group together, the Denen Dora through the KH-1 fossil assemblages group together, and the KH-2 submember fauna exhibits differences from the rest of the lower section. The KH-2 fossil assemblage contains several new species to the Hadar Formation with the first appearance of *Camelus*, *Pelorovis, Syncerus, Kobus sigmoidalis,* and a new genus of Antilopini. There are also species that have been recovered from older submembers that are not present in the KH-2 submember (Table 8).

Discussion

Habitat reconstructions

Basal Member. No hominins have been recovered from this member at the Hadar Site, but are present from this time period across the Awash River at Dikika (Alemseged et al., 2005). The Basal Member at Hadar encompasses ~30,000 years (~3.42–3.45; Campisano, 2007) and for this study includes the fossils recovered from the Sidi Hakoma Tuff (SHT). The fauna from the Basal Member indicates that the habitat was likely a mosaic of woodland and shrubland based on the position of the submember in the correspondence analysis near Hluhluwe and Sudan Woodland (Fig. 5). Abundances of bovids also indicate both open areas (alcelaphins) and more wooded areas (tragelphins). The abundance of Aepycerotini is also high indicating lightly wooded areas (Skinner and Smithers, 1990; Fig. 8). Analyses of pollen (Bonnefille et al., 2004) suggest that deposits directly below the SHT are the most forested of their samples. The pollen collection for this member was taken from a single locality and possibly sampled a forested area near a channel. The depositional environment begins with a lacustrine phase followed by a deltaic channel in a flat plains environment (Campisano and Feibel, in press).

Sidi Hakoma Member. The SH-1 and SH-2 were deposited between ~3.42–3.35 Ma (70,000 years), while SH-3 and SH-4 were deposited between ca. 3.35 and 3.26 Ma (94,000 years) (Campisano, 2007). The SH-1, directly above the SHT, is an ecotone habitat as evinced by the adaptations of the taxa creating the position in the CA (Figs. 5 and 7). In these analyses, the mammalian adaptations indicate higher annual rainfall and less seasonal environments than found in any other Hadar submember. This habitat reconstruction is supported by data from

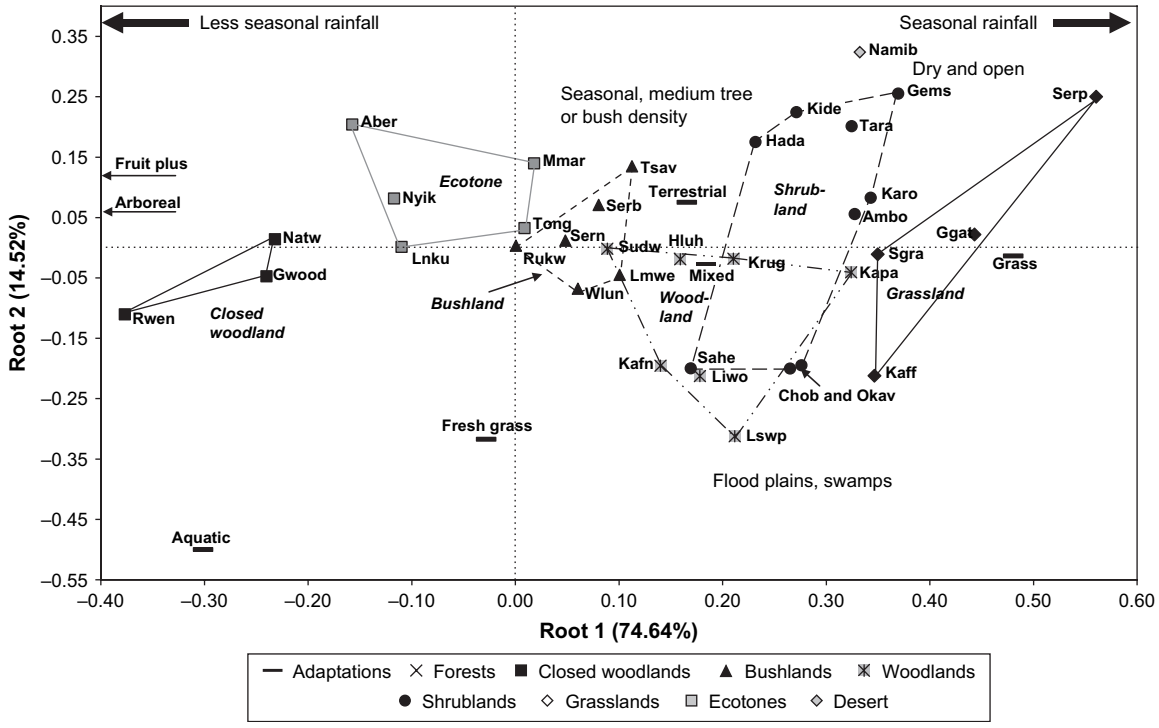


Fig. 4. Correspondence analysis of significantly different mammalian adaptations and modern habitat types without modern forests pictured. This is a figure of the same correspondence analysis as presented in Fig. 3, but modern site abbreviations are included and forests are removed to make the plot easier to read. Most of the woodlands fall below the origin of the y-axis indicating that they are wetter, in general, than many of the bushland habitats. Bushland habitats tend to have poorer soils than those of woodlands (White, 1983). Abbreviations for the sites can be found in Table 1 and further descriptions of the adaptations are located in Table 2.

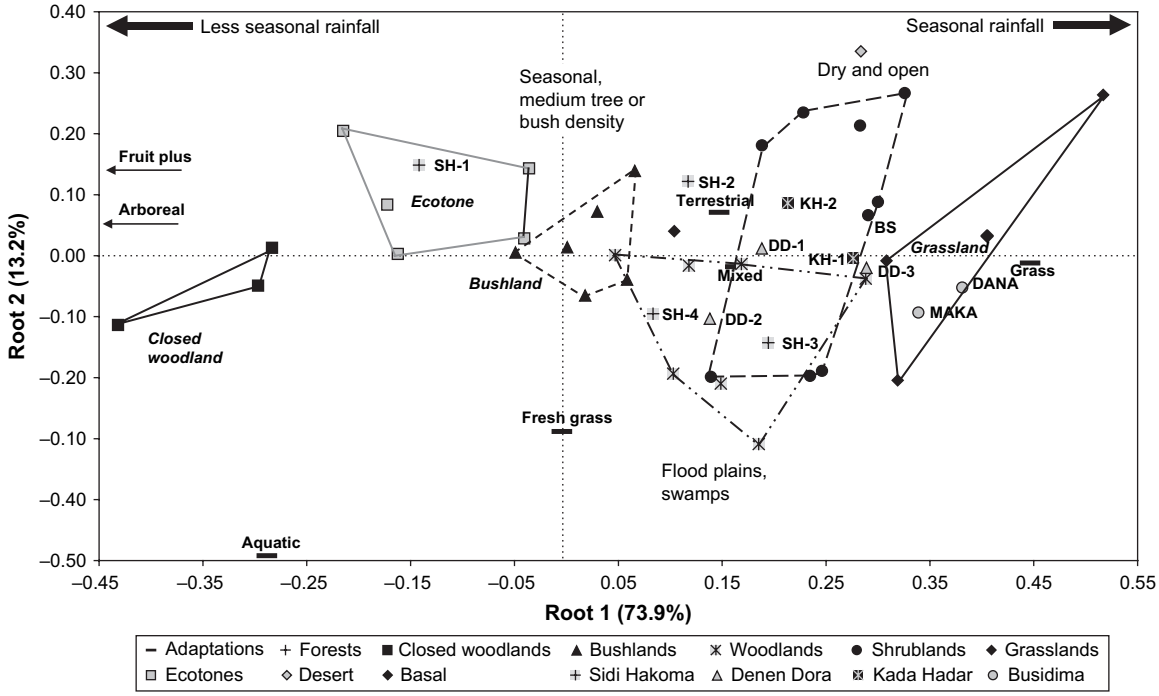


Fig. 5. Correspondence analysis of significantly different mammalian adaptations from extant sites and Hadar submembers without modern forests illustrated. The results are the same as those depicted in Figs. 3 and 4 with the Hadar submembers distributed, for the most part, in the region of intermediate habitats. The SH-1 submember falls within the ecotone habitats; the Makaamitalu and Danauli members are positioned with the more open grasslands. Abbreviations for the Hadar sites can be found in Table 1; modern sites are unlabeled for visual clarity; further descriptions of the adaptations are located in Table 2.

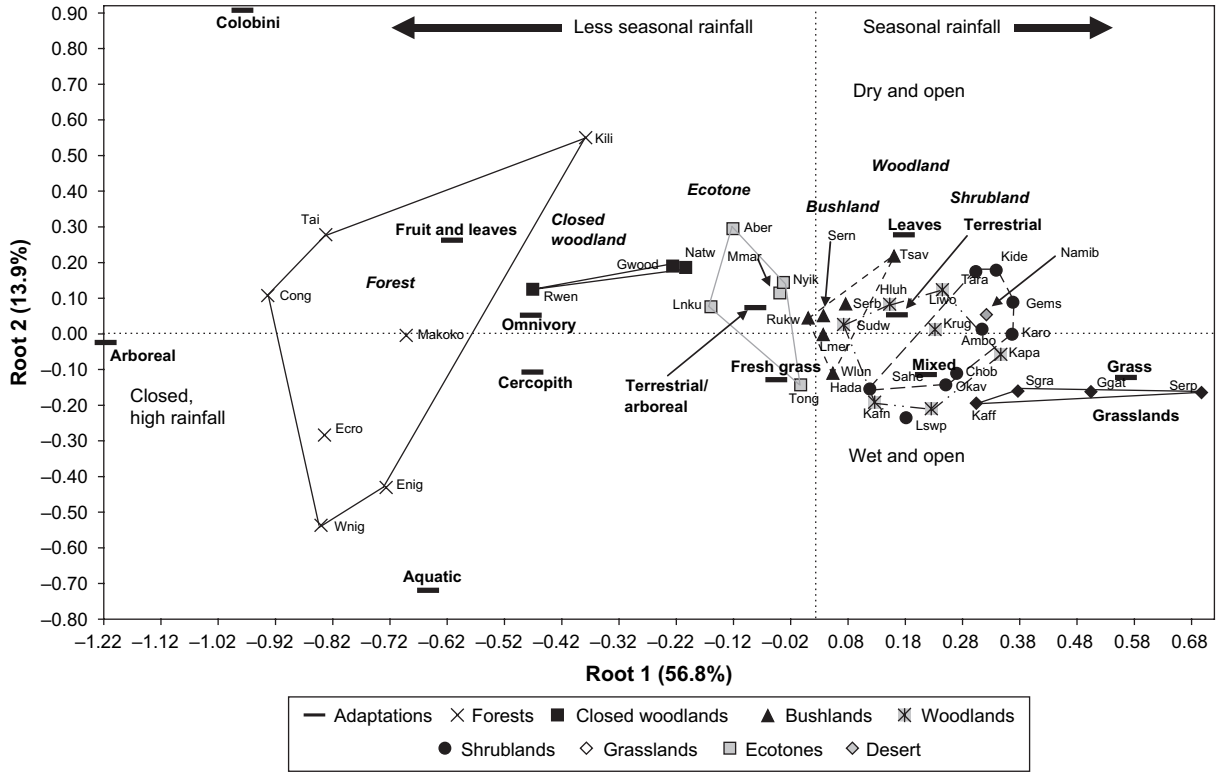


Fig. 6. Correspondence analysis of modern sites based on species from the selected orders dataset: Artiodactyla, Perissodactyla, Proboscidea, and Primates. Although there is still a wet, closed to dry, open gradient along both the x- and y-axes, the discrimination among the intermediate cover habitats has much less resolution. In addition, various adaptations, such as browsing and the presence of colobines, affect the distribution of the sites such that the habitats that have high numbers of these species group together, rather than grouping together by habitat. Abbreviations for the sites can be found in Table 1 and further descriptions of the adaptations are located in Table 2.

Hailemichael's (2000) analysis of oxygen and carbon isotopes of mollusks that indicate only a three month dry season during these deposits. The SH-1 submember also has low proportions of alcelaphins and antilopins (Fig. 8), although the minimum numbers of bovid individuals for this submember are quite low (Table 7). The SH-2 unit is reconstructed as woodland-shrubland, similar to the Basal Member. The pollen record from the SH-2 (Bonnefille et al., 2004) shows that this submember may have been more forested toward the top of the sequence. However, pollen also indicates a mosaic, changing environment over the estimated 20,000 years of deposition from which pollen was sampled. The SH-3, encompassing ~50,000 years (Campisano, 2007) is reconstructed as scrub woodlands and grasslands and, due to the presence of reduncins and hippoptamids, these grasslands may have existed in floodplains or wetland habitats (e.g., similar to Chobe National Park and the Okavango Delta). The SH-3 bovids all have similar abundances (Fig. 8) and its position in Fig. 9 is close to the origin, also indicating that there are no high proportions of individuals in any bovid tribe. The SH-4 submember spans ~44,000 years, and is similar in habitat to the SH-3, although the percentage of reduncins is higher (Fig. 8). Reduncins possibly indicate an expansion of lake margin wetlands. The percentage of *A. afarensis* drops below 5% in the SH-4 (and subsequent DD-1) submembers, possibly because of the extent of the lake in the region during these times. The SH-4 begins the most significant lake transgression episode of the Hadar Formation (Campisano and Feibel, in press).

The pollen data from the upper Sidi Hakoma is discontinuous but suggests drier environments with 800–900 mm of rainfall per annum, and has been interpreted as wet grassland (Bonnefille et al., 2004). These rainfall figures indicate closed to open woodland vegetation (Table 2), and there is also the influence of the lake transgression that would likely affect vegetation patterns from the lower part of the Sidi Hakoma Member. Hailemichael (2000) showed that there were changes in the environmental conditions in which mollusks developed between ~3.34 Ma and ~3.28 Ma and again between ~3.28 and ~3.22 Ma that may have been the result of transgression and regression of Lake Hadar into the region. That is, transgression of the lake into the Hadar area would occur with greater rainfall. This mirrors the change from wetter to drier conditions suggested by the mammalian fauna in this member. The depositional character of the Sidi Hakoma Member shows changes from channel deposits to fluvial and flood deposits within a large meandering river system (Campisano and Feibel, in press).

Denen Dora Member. The entire Denen Dora Member encompasses ~56,000 years with sediment deposition in each successive submember decreasing such that the least time is in the DD-3 submember (Campisano, 2007). The DD-1 fauna shows similarities to woodlands such as Hluhluwe and Kruger

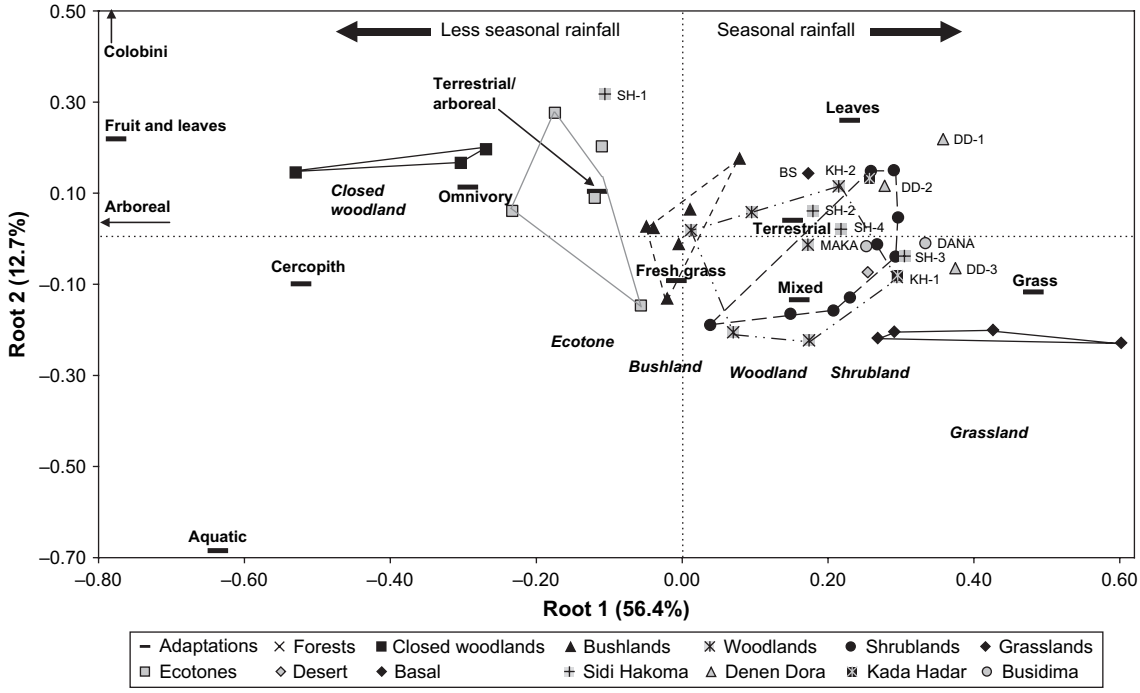


Fig. 7. Correspondence analysis of modern sites and Hadar submembers based on species from the selected orders dataset: Artiodactyla, Perissodactyla, Proboscidea, and Primates. The results are relatively the same as those in Fig. 6. Hadar submembers are affected by the presence of a number of species with the browsing (leaves) adaptation and by the presence of colobines. There are very few colobines found outside of forests (or riverine forests) in modern habitats. Therefore, this is an interesting examination of the community structure of various sites, including the Hadar submembers, but the selected orders database is not as good as the other database (mammals >4 kg) at reconstructing habitats. Abbreviations for the Hadar sites can be found in Table 1; modern sites are unlabeled for visual clarity; further descriptions of the adaptations are located in Table 2.

National Parks in South Africa, both of which are mosaics that include gallery forest and bushland in overall woodland habitats (Fig. 5). Geological data suggests that the silts and clays in the DD-1 submember were deposited under the most extensive transgression of a shallow lake (Aronson and Taieb, 1981; Campisano and Feibel, in press), but much of the fauna above this level derives from the sands and therefore is more representative of a fluvial or deltaic depositional environment. Although little fossil material has been recovered from the DD-1 submember, there is a relative increase in reduncins, aepycerotins, and tragelaphins compared with the Sidi Hakoma Member, and the fauna supports an overall woodland habitat reconstruction (Figs. 8 and 9). In contrast, the adaptations of the taxa from DD-2 submember indicate a habitat that was intermediate between modern lacustrine habitats (as represented by Lake Mweru and West Lunga National Park), and more woodland wetland habitats (similar to Kafue and Liwonde National Parks) (Fig. 5). A wetland or floodplain habitat is also supported by the numbers of Reduncini recovered from the DD-2 submember, and it is the only submember located in the same quadrant with the reduncins emphasizing its distinction (Fig. 10). Percentages of *A. afarensis* increase in this submember.

Analyses of the DD-3 fauna indicate a seasonal rainfall regime and open woodland or wooded grassland (similar to Kapama Game Reserve and the Southern Savanna Grasslands

Table 6
Number of shared species in each pair of Hadar submembers

	BS	SH-1	SH-2	SH-3	SH-4	DD-1	DD-2	DD-3	KH-1	KH-2	MAKA	DANA
BS	20											
SH-1	13	21										
SH-2	13	12	21									
SH-3	14	12	12	21								
SH-4	14	12	13	13	17							
DD-1	9	8	9	10	11	13						
DD-2	12	10	11	16	12	12	26					
DD-3	11	10	10	13	11	12	17	23				
KH-1	10	9	10	12	11	10	13	15	23			
KH-2	10	9	11	13	9	9	17	14	15	34		
MAKA	1	1	0	1	1	2	2	2	3	7	24	
DANA	0	0	1	1	0	1	2	1	2	7	15	20

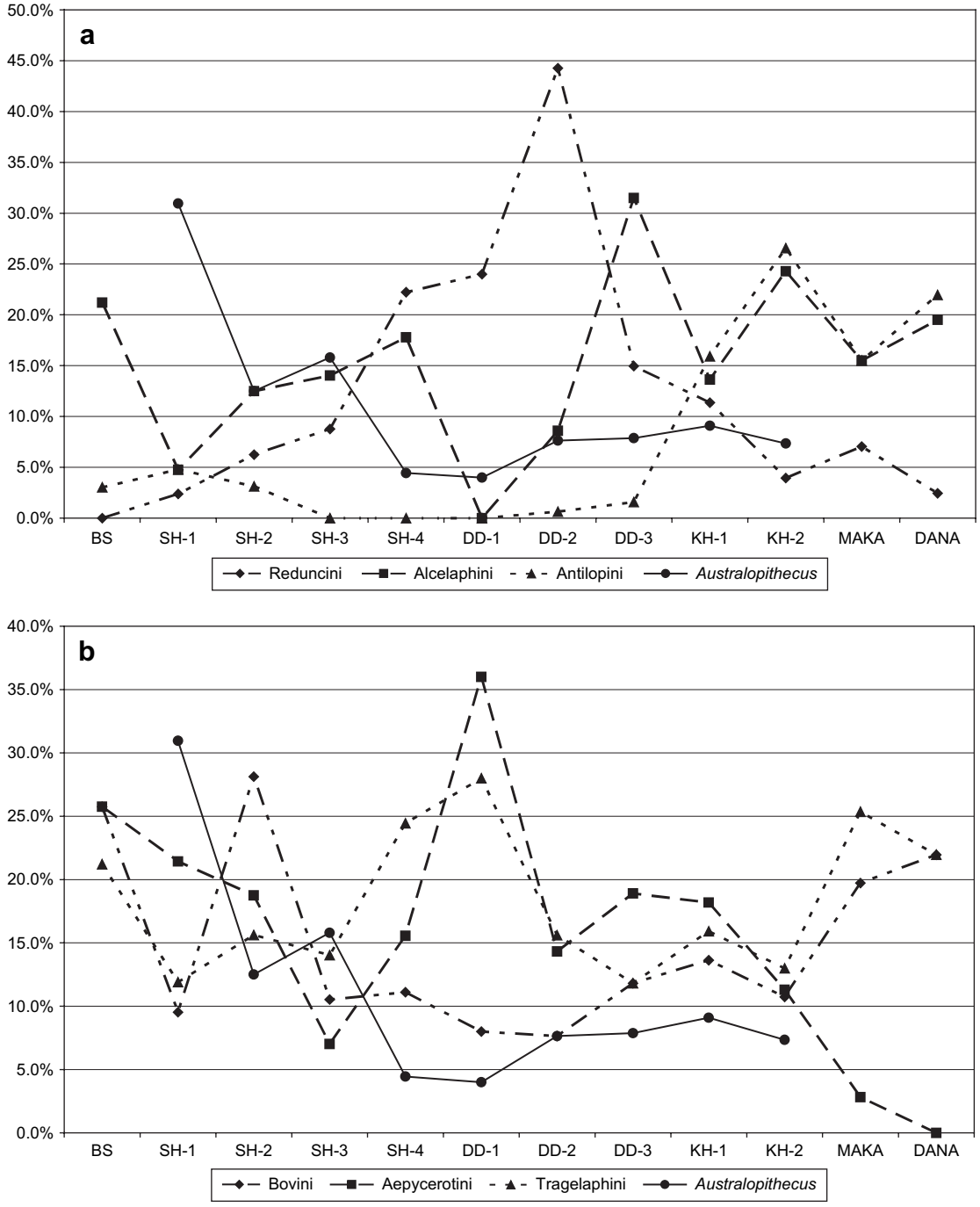


Fig. 8. a) Distribution across Hadar submembers of the minimum numbers of individuals in the bovid tribes Antelopini, Alcelaphini, and Reduncini with *Australopithecus afarensis*. (b) Distribution across Hadar submembers of the minimum numbers of individuals in the bovid tribes Aepycerotini, Bovini, and Tragelaphini with *Australopithecus afarensis*. Abbreviations for the Hadar sites can be found in Table 1. See text for discussion.

of South Africa; Fig. 5). The grassland reconstruction is supported by the substantial increase in alcelaphins, but as the reduncins are reduced it is likely this was a drier habitat than the DD-2 submember. Bonnefille et al. (2004) collected pollen only from the lower Denen Dora Member (equated here to DD-1), which indicates rainfall of about 800–900 mm per year. In contrast to the upper Sidi Hakoma sample, pollen indicates that the predominant vegetation of the Denen Dora (DD-1) was dry grassland, although there were sedges and reed pollen recovered as well (Bonnefille et al., 2004). The depositional environment of the DD-2 submember shows fluvial and deltaic channels, while the DD-3 shifts to a large scale meandering river system with over bank flood deposits (Yemane et al., 1996; Campisano and Feibel, in press).

Kada Hadar Member. There are two collection units that encompass the ~3.2–2.94 Ma timespan, divided by the Bouroukie Tuff 1 (BKT-1) at ca. 3.12 Ma (Campisano, 2007). The

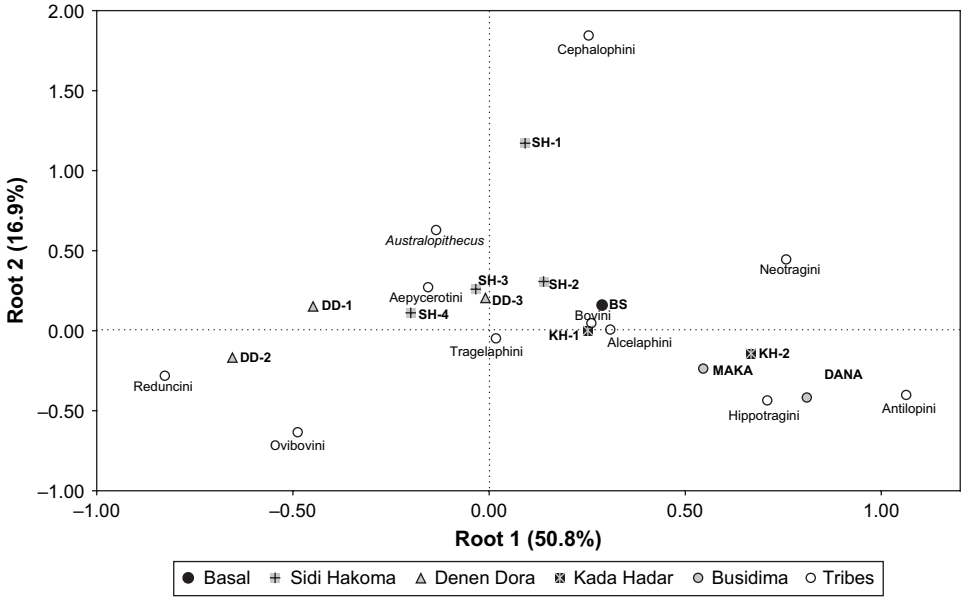


Fig. 9. Correspondence analysis of the minimum numbers of individuals in bovid tribes and *A. afarensis* in Hadar submembers. Those tribes that cluster around the origin are found in more of the Hadar submembers than those that are located in the periphery of the graph. Those that are found in the outer limits, such as Antilopini, are found in high abundance in one or two submembers or those, such as Cephalophini, that are rare and found only in one submember. Abbreviations for the sites can be found in Table 1; abbreviations for the adaptations are in Table 2.

KH-1 fauna most closely resembles that from open woodland with some edaphic grassland (similar to Amboseli National Park; Fig. 5). The abundance of antilopins increases, but the alcelaphins actually are reduced from the DD-3 submember (Fig. 10). The depositional environment at the base of this submember was likely a shallow lake, while fluvial deposits appear towards the top of the collection unit. The KH-2 fauna reveals a habitat that is relatively dry and open (similar to Amboseli National Park and the Nama Karoo; Fig. 5). The KH-2 fauna also displays high proportions of Antilopini and Alcelaphini bovids, which indicate drier environments (Vrba, 1980b), especially when contrasted with other Hadar submembers (Figs. 9 and 10). On the other hand, pollen data suggest a habitat mosaic with a return to more humid, woodland conditions (Bonnefille et al., 2004). As mentioned above, studies of landscape ecology for each member are currently underway and this may reveal regions of more closed physiognomies within the KH-2 collection unit (Campisano and Reed, 2007). However, the broad-based habitat classification for the KH-2 is open woodland or shrubland. The geology of the KH-2 submember suggests floodplain deposits with seasonal vertisols interspersed with fluvial/channel sequences with a transition to a lacustrine phase during BK-2 times (Campisano and Feibel, in press).

Busidima Formation. The reconstructed habitat for the Makaamitalu Basin, from which *Homo* aff. *H. habilis* was recovered, is similar to a physiognomy between the Kafue Flats and the southern savanna grassland (Fig. 5). The high percentage of tragelphins, however, may indicate wooded grassland or more closed patches of trees. The high percentages of bovins likely indicate tall grassland such as are found in wetlands. Sediments suggest overbank floodplain deposits which are

Table 7
Numbers of individuals of each bovid tribe and *Australopithecus* in each Hadar submember

Species	BS	SH-1	SH-2	SH-3	SH-4	DD-1	DD-2	DD-3	KH-1	KH-2	MAKA	DANA	Total
Aepycerus	17	9	6	4	7	9	45	24	8	20	2	0	151
Alcelaphini	14	2	4	8	8	0	27	40	6	43	11	8	171
Antilopini	2	4	1	0	0	0	2	2	7	47	11	9	85
Bovini	17	4	9	6	5	2	24	15	6	19	14	9	130
Cephalophini	0	2	0	0	1	0	0	0	0	0	1	0	4
Hippotragini	2	0	1	1	1	0	2	0	1	3	5	4	20
Neotragini	0	2	0	0	0	0	0	2	0	2	3	1	10
Ovibovini	0	0	0	0	0	0	2	0	0	0	1	0	3
Reduncini	0	1	2	5	10	6	139	19	5	7	5	1	200
Tragelaphini	14	5	5	8	11	7	49	15	7	22	18	9	170
Australopithecus	0	13	4	6	2	1	24	10	4	13	0	0	77
	66	42	32	38	45	25	314	127	44	176	71	41	1021

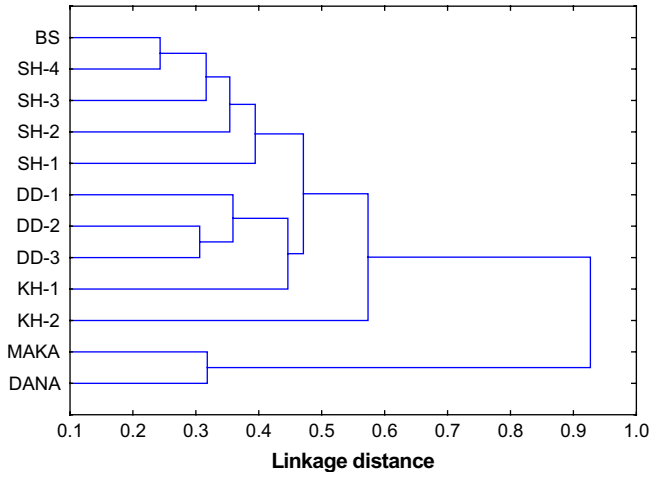


Fig. 10. Dendrogram of the relationship of the presence and absence of all taxa (irrespective of body size) among submembers at Hadar. The submembers tend to group from top to bottom by older to younger deposits. As shown in Table 7, the change of species in the KH-2 submember causes it to cluster next to all of the other submembers in the Hadar Formation. The two basins from the Busidima Formation have fossil faunas that are almost completely different from the Hadar Formation faunas, sharing only seven species with all Hadar Formation submembers combined.

consistent with the fauna recovered. The Danauli drainage, ~2.0 Ma is unique from other reconstructed Hadar habitats in having the highest number of grazers, as well as abundant fresh grass grazers and aquatic mammals, all of which indicate both open wet grasslands and floodplains (Fig. 5).

Species ecology and changes through time

There are minor fluctuations in habitat physiognomy through time at the Hadar hominin site. Changes through time also reflect spatial patterns on the landscape. Walther's Law states that vertical facies associations observed in the stratigraphy reflect horizontal landscape associations during the times of depositions (Middleton, 1973). Because this implies that all of the habitats existed, but shifted into and out of the Hadar research area spatially through time, there might be few species extinctions expected. This type of sequence through time should reflect the presence of eurytopic species in most collection units. Stenotopic species may appear intermittently, migrating into and out of the area as they track preferred habitats. Taxonomic similarity among members should reflect this pattern with all collection units being fairly uniform in membership. Abundances in bovid tribes can also be representative of shifting habitats across the landscape.

The slight differences in species composition between collection units could be due to small changes in habitat as discussed above, or could be the result of a taphonomic bias wherein particular taxa from each member were not recovered. There are several species present in each submember, Basal to KH-2. These taxa are listed in Table 8. *Tragelaphus* aff. *T. kyaloae* is present in all but the SH-1 and KH-2, which are the most closed and the most arid habitats, respectively, below the disconformity. As these taxa are present in all or most of the fossil assemblages, it is reasonable to suggest that they

Table 8
Appearances and disappearances of species in the Hadar Formation

Taxa present in Members BS - KH-2	Changes in taxa in KH-1 and KH-2	New taxa in the MAKA and DANA	Disappearances in the MAKA and DANA
Australopithecus afarensis (at Dikika/Basal)	*Parmularius* cf. *P. altidens*	*Aepyceros* sp.	*Damalops denendorai*
Aepyceros sp. nov.	*Gazella* cf. *G. vanhoepeni*	*Connochaetes* sp.	*Parmularius* cf. *P. altidens*
Ugandax coryndoni	*Oryx* sp.	*Damaliscus* sp.	*Ugandax coryndoni*
Kolpochoerus afarensis	*Eurygnathohippus afarense*	*Gazella praethomsoni*	*Praedamalis deturi*
Notochoerus euilus		*Kobus ellipsiprymnus*	*Tragelaphus kyaloae*
Eurygnathohippus hasumense	**Disappearances in the KH-1**	*Redunca* sp.	*Tragelaphus nakuae*
Theropithecus darti	*Kobus oricornis*	*Tragelaphus gaudreyi*	*Tragelaphus* cf. *T. pricei*
Damalops sp.*	*Nyanzachoerus kanamensis*	*Tragelaphus scriptus*	*Giraffa jumae*
		cf. *Hippopotamus amphibius*	*Giraffa pygmaea*
	New taxa in the KH-2	*Metridiochoerus modestus*	*Giraffa stillei*
	Gazella janenschi	*Metridiochoerus andrewsi*	*Kolpochoerus afarensis*
	Antilopini Gen et sp. nov.	*Civettictus*	*Notchoerus euilus*
	Pelorovis sp.	*Equus* sp.	*Eurygnathohippus hasumense*
	Syncerus acoelotus	*Cercopithecoides kimuei*	*Eurygnathohippus afarense*
	Kobus sigmoidalis	*Theropithecus oswaldi*	*Theropithecus darti*
	Tragelaphus nakuae	*Homo habilis*	*Deinotherium bozasi*
	Tragelaphus cf. *T. pricei*	*Millardia coppensi*	*Australopithecus afarensis*
	Camelus sp.	*Thryonomys swinderianus*	
	Kolpochoerus limnetes		
	Hyaena makapani		
	Disappearances in the KH-2		
	Alcelaphini sp. large		
	Kobus sp. nov.		
	Hippopotamus afarensis		

* *Damalops sidihakomai* and *D. denendorai* are an evolutionary lineage (Vrba, 1997).

are relatively eurytopic species. This includes, of course, *A. afarensis*.

The clusters in the dendrogram also show successive time groups such that a gradual change of species is indicated. The Basal and Sidi Hakoma member fauna group together and the Denen Dora Member and KH-1 submember group together, which is mirrored by the change to significant wetland habitats and the presence of large numbers of Reduncini. As a result of the addition of many new taxa and the loss of some others, there was a substantial species change between the KH-2 and the group that encompasses the Basal to KH-1 submembers (Table 8; Appendix 1). Along with the replacement of *A. afarensis* by *Homo* aff. *H. habilis*, there was almost a complete change in mammalian fauna between ~2.94 Ma and the sites younger than ~2.35 Ma, the Makaamitalu and Danauli. Taxa that are shared between these younger deposits and some of the older Hadar members include *Elephas recki* (although these are different subspecies), *Sivatherium maru-sium,* and *Crocuta dietrichi*. The KH-2 fossil assemblage shares these three taxa with the Makaamitalu and Danauli basins, as well as specimens of *Syncerus aceolotus, Kolpochoerus limnetes*, and *Hyaena makapani*.

Species endemism and migrations

Several fossil species have only been recovered from the Hadar region, or from the northern and central Afar, (e.g., Gona and Middle Awash): *Trilobophorus (*aff. *Hippopotamus) afarensis* and *Hexaprotodon (*aff. *Hippopotamus) coryndoni*, hippopotamid species (Boisserie, 2005); *Aepyceros* sp. nov., *Kobus* sp. nov., *Tragelaphus* sp. nov., (Gentry, 1981); *Ugandax cornydoni* (Gentry, 2006), *Damalops ''sidihakomai''*, *D. ''denendorai''* (Vrba, 1997), and *Budorcas churcheri* (Gentry, 1996), all bovid species; *Homotherium* sp. nov.; two new species of *Enhydriodon*, carnivore species (L. Werdelin, pers comm.); and *Eurygnathohippus afarense*, an equid (Bernor et al., 2005). Gentry (1981) suggested that the *Aepyceros* and *Kobus* species at Hadar were ancestral to later taxa present in the Shungura Formation, although it is difficult to identify origination of taxa with a spotty fossil record. The species composition throughout the Hadar depositional period provides a taxonomically unique community in which these populations of *A. afarensis* existed.

The species composition of the KH-1 and KH-2 submembers, compared with the fossil assemblages of older submembers, may indicate climatic or other changes that allowed species migrations into the area, as well as possible speciation and extinction events. A few species appear for the first time at Hadar in the KH-1 submember, having been recovered earlier in time elsewhere: *Gazella praethomsoni*, *Antidorcas* (Turkana Basin; Behrensmeyer et al., 1997), and *Parmularius* sp. [FAD 2.85 Ma in the Turkana Basin, but present in Africa from 3.76 Ma (Vrba, 1995)]. The KH-2 fossil assemblage has a new bovid species that has not been recovered elsewhere (i.e., a new genus and species of Antilopini). Additionally, there are taxa from the KH-2 submember that have been recovered previously from older Turkana Basin localities: *Camelus*, *Tragelaphus nakuae*, *Tragelaphus* cf. *T. pricei*, *Pelorovis,* and *Kobus sigmoidalis* (Harris, 1991; Behrensmeyer et al., 1997; Bobe and Eck, 2001). *Syncerus aceolotus* has also been recovered from the KH-2 submember, but the first appearance datum (FAD) for this taxon in the Turkana Basin is ~2.85 Ma.

In addition, abundances of many of the animals in the Kada Hadar Member are changed dramatically from those recovered from the Denen Dora Member. Thus, the high abundances of antilopins and some of the other arid adapted taxa, not present in the older fossil assemblages at Hadar, likely indicate a migration of mammals into the area. This migration was either preceded or accompanied by a change in vegetation physiognomy.

Australopithecus afarensis *habitats*

Australopithecus afarensis has been recovered from all of the collection units below the BKT-2 and as such appears not to have had a narrow habitat preference. Fluctuations in habitats through time at Hadar were likely caused by abiotic changes, such as shifts in hydraulic systems across the landscape caused by both climatic alterations and geological activity (Campisano and Feibel, in press). The patterns of meandering and braided streams changed, as well as the lateral extent of lacustrine environments due to these factors. These in turn influenced slight vegetation changes and pattern on the landscape. Under these conditions, it is feasible that spatial movements of habitats across the landscape are being detected in this study. Over 500,000 years, it is probable that habitats in the Hadar research area moved completely out of the 120 km^2 area, with subsequent reduction of mammals adapted to those particular habitats. For example, some mammals are associated with floodplains and edaphic grasslands and consequently track these habitats across the landscape. The KH-2 collection unit is depauperate in all of the floodplain adapted mammals that existed in high abundances throughout much of the Denen Dora Member. These species may have moved eastward to follow favored habitats, or become extinct. Only research in areas outside of Hadar will answer this question.

Slight changes in habitats are supported by palynological findings (Bonnefille et al., 2004), as well as by paleosol and molluskan carbon and oxygen isotope data from Hadar. Hailemichael (2000) also found changes between wetter and drier habitats such as those between open and closed woodlands. Analyses of paleosols indicate that there were fewer than 40% C_4 grasses in the entire lower section of the Hadar Formation (below the BKT-2). As there is no way to differentiate C_4 wetland grasses (e.g., sedges) from other grasses (e.g., *Themeda*), it is possible that the grasses detected by Hailemichael in much of the lower Hadar Formation are mainly from wetlands derived from over bank flood deposits and shallow lakes. Hailemichael (2000) suggested that the grass pollen discovered by Bonnefille et al. (1987) is from cattails because the pollen was recovered from a time of extensive shallow lake transgression. Bonnefille et al. (2004) showed the presence of sedges and other water adapted grasses that mirror this

suggestion by Hailemichael. The paleosol data also indicate fluctuating intermediate woodland type habitats with no discernible trend toward open secondary grasslands or more arid habitats for much of the Hadar Formation. Finally, Bobe and Eck (2001) compared bovid tribal representation at Hadar with those of the Shungura Formation. They conclude, as well, that there is no trend to more open arid habitats as there is in the Omo and that the overall habitat at Hadar is more open woodland and grassland than the more closed vegetation of the Shungura.

The Laetoli sample of *A. afarensis* has been recovered with other mammals that indicate fairly closed woodland (Andrews, 1989; Reed, 1997) or woodland with patches of forest (Andrews, 2006). Other researchers have suggested that due to the absence of aquatic species and the presence of numerous dik-diks (*Madoqua aviflumens*) and Alcelaphini at Laetoli that a more open, arid habitat is indicated (Harris, 1987). Evidence from mollusks (Verdcourt, 1987) suggests that the mean annual rainfall at the time ranged between 625 and 1,000 mm. Today, this rainfall regime also supports closed woodlands through bushlands. In addition, the presence of *Paracolobus* and *Galago* indicate that there must have been some tree cover. As there have been no aquatic mammals recovered from the Upper Laetolil Beds thus far, this suggests that these primates and small antelopes were not living in a riverine forest. Harrison (2005) reported that fossil bird species recovered from Laetoli indicate some bushland thickets and Andrews (2006) suggested there are patches of forest across the landscape. Kingston and Harrison (2007) show that most of the fauna sampled from the Upper Laetolil Beds are browsers and mixed feeders and also indicate the presence of trees and bushes. Thus, the earliest known *A. afarensis* material was found in deposits showing habitats in which trees and or bushes were fairly plentiful. It is also interesting to note that while the deposits of *A. afarensis* at Laetoli and Hadar share some perissodactyls, giraffids, suids, and proboscideans, the bovid taxa and those primates other than *A. afarensis* are not very similar.

A. afarensis has also been recovered from Maka, Ethiopia (White et al., 1993). These authors suggest that associated fauna indicate a habitat that is similar to that of the upper Denen Dora Member (assumed to be the DD-3). Bushland or woodland with floodplain grassland is likewise indicated for the Maka locality. The Tulu Bor Member of the Koobi Fora Formation overlaps the entire Hadar Formation (Basal - KH-2) and has yielded a few *A. afarensis* specimens (Kimbel, 1988). Feibel et al. (1991) described the depositional environment during Tulu Bor times as fluvial with floodplain lakes. Harris (1991) suggested that the habitat of Koobi Fora at this time was gallery forests amid floodplains, wet grasslands, and woodlands. According to these authors, both the depositional environment and the reconstructed habitats of the Tulu Bor Member of Koobi Fora were similar to Hadar. However, the habitats of Hadar, based on mammalian adaptations, were wetter and possibly more closed than those of the Tulu Bor Member. Reconstruction of the habitat during the Tulu Bor Member, based on published faunal lists is that of shrubland with wetlands and floodplains (Reed, 1997). In addition, camels and other fauna similar to that of the KH-2 collection unit have been recovered from the Tulu Bor Member. As there are only published faunal lists for the entire Tulu Bor Member, it may be that the faunal changes are similar to Hadar and these more arid-adapted mammals arrive towards the top of the member. Compared with Hadar, however, there are few *A. afarensis* specimens from the Tulu Bor Member, and this may also indicate that the habitat of the entire member was somewhat different from Hadar.

White et al. (1993) suggested broad habitat tolerance for *A. afarensis,* and indeed, the species has thus far been recovered from regions in which the reconstructed habitat ranges from closed woodland through more open, but wet woodland and shrubland. There is no direct evidence that *A. afarensis* only existed in riverine forests or grassland habitats, or that they preferred one habitat over another. It is tempting to equate the aridification in the Kada Hadar Member with the extinction of *A. afarensis*. However, sediments at Hadar are sparse or missing altogether from ~2.90–2.35 Ma thus obscuring details of the species' demise. All that can be said is that they are no longer present at 2.35 Ma and most of the fauna, including hominins, has been replaced.

Conclusions

The ancient habitats of the Hadar hominin site are reconstructed as physiognomies ranging, for the most part, in the intermediate cover habitats of bushland, open woodland, and shrubland with varying regions of wetlands or edaphic grasslands through time. There is no trend in time progression of wet, closed habitats to those that are dry and open at Hadar, although there may be a gradual drying trend amid fluctuations. There are species present in all of the submembers of the Hadar Formation that appear to be ecologically eurytopic, including *A. afarensis*. The changes in habitat and the species representations at Hadar may well represent responses to local and or regional climate fluctuations and local geological events. There is a change in overall species composition at ~3.0 Ma, which appears to signify an ecological change. This change in species may be associated with geological events that allowed migration of species into the area, although this hypothesis needs to be tested further. Overall, *A. afarensis* survived in a variety of habitats and was likely a eurytopic species. This does not necessarily mean that *A. afarensis* was catholic in its diet, as preferred food items may have been obtainable in many different environments.

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Formation. Some of these data are published (e.g., Gentry, 1981; Vrba, 1997; Frost and Delson, 2002; Boisserie, 2005; Geraads, 2005) and some of the work is ongoing. I benefited from the assistance of Lars Werdelin, Elisabeth Vrba, Amy Rector, Charles Lockwood, Margaret Lewis, Denis Geraads, Steve Frost, Eric Delson, Rene Bobe, Ray Bernor, and Zeresenay Alemseged. The Authority for Research and Conservation of Cultural Heritage in the Ministry for Youth, Sports and Culture of the government of Ethiopia have been helpful during both field and museum research done there. Thanks to Gerry Eck, Lillian Spencer, Chris Campisano, Bill Kimbel, Gregg Gunnell, Peter Andrews, and an anonymous reviewer who all read early versions of this manuscript. Thanks also go to Lillian for keeping me on track despite numerous obstacles. This research was supported by the L.S.B. Leakey Foundation and the Institute of Human Origins.

Appendix 1. Presence of fauna is indicated by an X; absence is indicated by 0. Abbreviations for trophic adaptations are as follows: B = leaves, C = meat, CB = meat/bone, CI = meat/invertebrates, FG = fresh grass, FL = fruit plus, G = grass, I = insects, MF = leaves and grass, OM = omnivorous, R = roots, bulbs. Abbreviations for substrate use include: A = arboreal, AQ = aquatic, F = fossorial, TA = terrestrial/arboreal, T = terrestrial. Further descriptions of each category can be found in Table 2.

	Trophic	Substrate	BS	SH-1	SH-2	SH-3	SH-4	DD-1	DD-2	DD-3	KH-1	KH-2	MAKA	DANA
Artiodactyla														
Bovidae														
Aepycerotini														
Aepyceros sp. nov.	MF	T	X	X	X	X	X	X	X	X	X	X	0	0
Aepyceros sp.	MF	T	0	0	0	0	0	0	0	0	0	0	X	0
Alcelaphini														
Connochaetes sp.	G	T	0	0	0	0	0	0	0	0	0	0	0	X
Damaliscus sp.	G	T	0	0	0	0	0	0	0	0	0	0	X	X
Damalops denendorai	G	T	0	0	0	0	0	0	X	X	X	X	0	0
Damalops sidihakomai	MF	T	X	X	X	X	X	0	0	0	0	0	0	0
cf. *Parmularius*		T	0	0	0	0	0	X	0	X	0	0	0	0
Parmularius cf. *P. altidens*	G	T	0	0	0	0	0	0	0	0	X	X	0	0
Parmularius sp. (small)	MF	T	0	[a]	[a]	0	0	0	0	0	0	0	0	0
Alcelaphini sp. (large)	G	T	X	0	X	X	X	0	X	X	X	0	0	0
Antilopini														
Antilopini species		T	X	X	0	0	0	0	0	X	0	0	0	0
Antidorcas species	MF	T	0	0	0	0	0	0	0	0	X	X	0	0
Gazella janenschi	MF	T	0	0	0	0	0	0	0	0	0	X	X	X
Gazella praethomsoni	MF	T	0	0	0	0	0	0	0	0	0	0	X	X
Gazella species	MF	T	X	0	X	0	0	0	X	0	0	0	0	0
Gazella cf. *G. vanhoepeni*	B	T	0	0	0	0	0	0	0	0	X	0	0	0
Gen. et sp. nov.	MF	T	0	0	0	0	0	0	0	0	0	X	0	0
Bovini														
Pelorovis sp.	G	T	0	0	0	0	0	0	0	0	0	X	X	X
Syncerus acoelotus	FG	T	0	0	0	0	0	0	0	0	0	X	X	X
Ugandax coryndoni	FG	T	X	X	X	X	X	X	X	X	X	X	0	0
?*Syncerus* sp.	?	T	0	0	0	0	0	0	0	0	X	0	0	0
Cephalophini														
Cephalophini sp.	FL	T	0	X	0	0	X	0	0	0	0	0	X	0
Hippotragini														
Hippotragini sp.	G	T	X	0	X	0	0	0	0	0	0	0	X	0
Oryx sp.	MF	T	0	0	0	0	0	0	0	0	X	X	X	X
Praedamalis deturi	G	T	0	0	0	X	0	0	X	0	0	X	0	0
Neotragini														
Madoqua avifluminis	FL	T	0	0	0	0	0	0	0	[a]	[a]	0	0	0
Raphicerus sp.	MF	T	0	0	0	0	0	0	0	X	0	0	X	0
Neotragini sp.		T	0	X	X	0	0	0	0	0	0	X	X	X
Ovibovini														
Budorcas churcheri	MF	T	0	0	0	0	0	0	X	0	0	0	0	0
Reduncini														
Kobus hadarensis	FG	T	0	0	0	X	X	X	X	X	X	0	0	0
Kobus sigmoidalis	FG	T	0	0	0	0	0	0	0	0	0	X	0	0
Kobus oricornis	FG	T	0	0	0	0	0	0	X	0	0	0	0	0
Kobus ellipsiprymnus	FG	T	0	0	0	0	0	0	0	0	0	0	X	X
? *Menelikia*	MF	T	0	[a]	[a]	[a]	0	0	0	0	0	0	0	0
Reduncini sp.	FG	T	0	X	X	0	0	0	X	0	X	0	X	0
Tragelaphini														
Tragelaphus gaudreyi	B	T	0	0	0	0	0	0	0	0	0	0	X	X
Tragelaphus grayi	B	T	X	X	[a]	[a]	0	0	0	0	0	0	0	0

Appendix 1 (*continued*)

	Trophic	Substrate	BS	SH-1	SH-2	SH-3	SH-4	DD-1	DD-2	DD-3	KH-1	KH-2	MAKA	DANA
Tragelaphus imberbis	B	T	0	0	0	0	0	0	0	0	0	0	X	0
Tragelaphus kyaloae	B	T	X	0	X	X	X	X	X	X	X	0	0	0
Tragelaphus nakuae	B	T	0	0	0	0	0	0	0	0	0	X	0	0
Tragelaphus pricei	B	T	0	0	0	0	0	0	0	0	0	X	0	0
Tragelaphus scriptus	B	T	0	0	0	0	0	0	0	0	0	0	X	0
Camelidae														
Camelus sp.	B	T	0	0	0	0	0	0	0	0	0	X	0	0
Giraffidae														
Giraffa jumae	B	T	0	0	X	X	0	0	X	0	0	X	0	0
Giraffa pygmaea	B	T	0	0	0	0	0	0	X	0	X	X	0	0
Giraffa stillei	B	T	X	X	0	X	0	0	X	0	0	0	0	0
Sivatherium marusium	B	T	0	0	0	0	0	X	X	X	0	X	X	X
Giraffa sp.	B	T	0	0	0	0	X	X	0	0	0	0	0	0
Hippopotamidae														
aff. *Hippopotamus coryndoni*[b]	FG	AQ	0	[a]	[a]	X	0	0	[a]	[a]	0	0	0	0
cf. *Hippopotamus amphibius*	FG	AQ	0	0	0	0	0	0	0	0	0	0	X	0
aff. *Hippopotamus afarensis*[b]	FG	AQ	X	0	X	X	X	0	X	X	X	0	0	0
Hippopotamidae sp.	FG	AQ	0	0	0	0	0	0	0	0	0	X	0	0
Suidae														
Kolpochoerus afarensis	OM	T	X	X	X	X	X	X	X	X	X	X	0	0
Kolpochoerus limnetes	OM	T	0	0	0	0	0	0	0	0	0	X	X	0
Metridiochoerus modestus	G	T	0	0	0	0	0	0	0	0	0	0	X	X
Metridiochoerus andrewsi	G	T	0	0	0	0	0	0	0	0	0	0	0	X
Notochoerus euilus	G	T	X	X	X	X	X	X	X	X	X	X	0	0
Nyanzachoerus kanamensis	G	T	X	X	X	X	X	0	0	cf	0	0	0	0
Carnivora														
Canidae														
Canis adustus	C	T	0	0	0	X	0	0	0	0	0	0	0	0
Canis sp.	C	T	0	0	0	0	0	0	0	0	0	X	0	0
Felidae														
Dinofelis petteri	C	TA	0	0	0	0	0	[a]	[a]	0	0	0	0	0
Felis serval	C	T	0	0	X	0	0	0	0	0	0	0	0	0
Panthera sp. (large)	C	T	0	0	X	0	0	0	0	X	0	X	X	0
Felidae sp.	C	?	0	0	0	0	0	0	X	0	0	0	0	0
Herpestidae														
Herpestidae sp.	?	?	0	0	0	0	0	0	0	0	X	X	0	0
Hyaenidae														
Chasmaporthetes cf. *C. nitadula*	CB	T	0	X	0	0	0	0	0	X	0	0	0	0
Crocuta deitrichi	C	T	0	0	X	X	0	0	X	0	0	X	0	X
Crocuta sp.	CB	T	X	X	0	0	0	0	0	0	0	0	0	0
Hyaena makapani	CB	T	0	0	0	0	0	0	0	0	0	X	0	X
Pliocrocuta perrieri	C	T	0	0	0	0	0	0	0	X	X	0	0	0
Hyaenidae sp.	C	T	0	0	0	0	X	0	0	0	0	0	0	0
Machairodontidae														
Homotherium sp. nov.	C	T	X	X	0	X	0	0	0	X	X	X	0	0
Megantereon sp.	C	TA	0	0	0	0	0	0	X	0	0	0	0	0
Machairodontidae sp.	C	T	0	0	X	0	0	0	0	0	0	0	0	0
Mustelidae														
Enhydriodon sp. nov. A	CI	AQ	X	0	0	0	0	0	0	0	0	0	0	0
Enhydriodon sp. nov. B	CI	AQ	0	X	0	0	X	0	0	0	0	0	0	0
Enhydriodon sp. nov. C	CI	AQ	0	0	0	0	0	X	X	X	0	0	0	0
Mellivora benfieldi	OM	F	0	0	0	0	0	0	X	0	0	X	0	0
Mustelidae sp.	?	?	0	X	X	0	0	0	0	0	0	0	0	0
Ursidae														
Ursid sp.	OM	F	0	0	0	0	0	0	0	X	0	0	0	0
Viverridae														
Civettictus	C	T	X	0	0	0	0	0	0	0	0	0	0	X
Lagomorpha														
Leporidae														
Leporidae sp.	MF	T	0	0	0	0	0	0	0	X	0	X	0	0
Perissodactyla														
Equidae														
Eurygnathohippus hasumense[c]	G	T	X	X	X	X	X	X	X	X	X	X	0	0
Eurygnathohippus afarense[c]	G	T	0	0	0	0	0	0	0	0	X	X	0	0

(*continued on next page*)

Appendix 1 (*continued*)

	Trophic	Substrate	BS	SH-1	SH-2	SH-3	SH-4	DD-1	DD-2	DD-3	KH-1	KH-2	MAKA	DANA
Equid sp.	G	T	0	0	0	0	0	0	0	0	0	0	X	X
Rhinocerotidae														
Ceratotherium	G	T	X	0	X	0	0	0	X	X	X	X	0	0
Dicerus	B	T	X	0	X	0	X	0	0	0	0	X	0	0
Rhinocerotidae sp.	B	T	0	0	0	0	0	X	0	0	0	0	0	0
Pholidota														
Manidae														
Manis sp.	I	T	X	0	0	0	0	0	0	0	0	0	0	0
Primates														
Cercopithecidae														
Colobini														
Rhinocolobus turkanensis	FL	TA	0	X	[a]	[a]	0	0	[a]	[a]	0	0	0	0
Cercopithecoides meavei	FL	TA	0	X	X	0	0	0	0	0	0	0	0	0
Cercopithecoides kimuei	FL	TA	0	0	0	0	0	0	0	0	0	0	0	X
Colobine sp.	FL	TA	X	0	0	0	0	0	0	0	0	X	0	0
Papionini														
Parapapio jonesi	FL	A	0	X	X	0	0	0	0	0	0	X	0	0
cf. *Parapapio*	FL	TA	0	X	0	0	0	0	0	0	0	0	X	X
Theropithecus darti	FL	TA	X	X	X	X	X	X	X	X	X	X	0	0
Theropithecus oswaldi	G	T	0	0	0	0	0	0	0	0	0	0	X	X
Cercopithecinae	FL	TA	X	0	0	0	0	0	0	0	0	0	0	0
Hominidae														
Hominini														
Australopithecus afarensis	FL	TA	0	X	X	X	X	X	X	X	X	X	0	0
Homo aff. *H. habilis*	OM	T	0	0	0	0	0	0	0	0	0	0	X	0
Proboscidea														
Deinotheridae														
Deinotherium bozasi	B	T	X	0	0	X	0	0	X	X	0	X	0	0
Elephantidae														
Elephas recki	MF	T	X	X	0	X	X	X	X	X	X	X	X	0
Loxodonta sp.	B	T	X	0	X	0	X	X	0	0	X	0	0	0
Rodentia														
Hystricidae														
Hystrix sp.	R	T	0	X	X	0	X	0	0	0	0	0	X	X
Xenohystrix crassidens	R	T	0	0	X	0	0	0	0	0	X	0	0	0
Muridae														
Golunda gurai			0	0	0	0	0	0	0	0	0	0	X	0
Millardia coppensi			0	0	0	0	0	0	0	0	0	0	X	X
Saidomys afarensis			0	0	0	0	0	0	0	X	0	0	0	0
Sciuridae														
Xerus rutilis			0	0	0	0	0	0	0	0	X	0	0	0
Spalacidae														
*Tachyoryctes pliocanes*is			0	0	0	0	0	0	0	X	X	X	0	0
Thryonomidae														
Thryonomys swinderianus	FG	AQ	0	0	0	0	0	0	0	0	0	0	X	X

[a] Indicates mammals not used in analyses for that submember, but shows distribution across possible submembers.

[b] Boisserie (2005) notes that aff. *Hippopotamus (Hexaprotodon) coryndoni* needs further cranial material for adequate species determination and that aff. *H. (Trilobophorus) afarensis* may be revised to a unique genus.

[c] Bernor et al. (2005) suggest these two species of hipparions at Hadar, although they also suggest that further analyses are necessary.

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